

BL54L15u

Introduction

Overview

This document describes key hardware aspects of the BL54L15u. This document is intended to assist device manufacturers and related parties with the integration of this radio into their host devices. Data in this document is drawn from several sources. For full documentation on the BL54L15u, visit:

<https://www.ezurio.com/bl54l15-micro>

General Description



Experience a new pinnacle of performance, efficiency, and security with our new BL54L15u series, built on Nordic Semiconductor's powerful **nRF54**

silicon. Elevating what you know and love from the nRF52 series, this next generation redefines Bluetooth LE and 802.15.4 solutions. Unleashing enhanced processing power, expanded memory, and innovative peripherals, the BL54L15u is the ultimate choice for low power connectivity, in the smallest footprint.

Powered by

Nordic's nRF54L15

(WLCSP 300um) SoC, our ultra compact BL54L15u modules deliver secure and robust Bluetooth LE and 802.15.4 with flexible programming via the Nordic SDK, Zephyr RTOS and

[Canvas Software Suite](#)

Featuring a

128MHz ARM Cortex M33

and

128MHz RISC-V coprocessor

, supported by 1.5 MB non-volatile memory and 256 KB RAM, the BL54L15u modules offer double the processing power (vs prior BL654 – nRF52840). The BL54L15u series brings out all nRF54L15 hardware features and capabilities including up to

+8 dBm

transmit power,

1.7V – 3.6V

supply considerations, and

NFC A-Tag

implementation.

It's further enhanced with state-of-the-art security and is designed for PSA Certified level 3. It supports services such as Secure Boot, Secure Firmware Update, Secure Storage plus protection from physical attacks.

Chat 

Note:

BL54L15μ hardware provides all functionality of the nRF54L15 chipset used in the module design. This is a hardware datasheet only – it does not cover the software aspects of the BL54L15μ. This is to acknowledge that information in this datasheet is referenced from the nRF54L15 datasheet.

This datasheet is subject to change. Please [contact Ezurio](#) for further information.

Application Areas

Building Automation
Security
Medical Peripherals
Industrial Sensors

Features & Benefits

The BL54L15μ device features and benefits are described in the following table.

Nordic nRF54L15

– 2.4 x 2.25mm WSCLP with 32 GPIOs utilized.

Multi-protocol support:

Bluetooth LE Core 6.2, 802.15.4 (Thread/Matter)

Cortex M33

processor core: 128 MHz ARM Cortex M33

RISC-V

co-processor core: 128 MHz VPR

Memory:

1.5MB non-volatile memory, 256 KB RAM

High Speed Peripherals

: - HS-SPI/UART, software defined peripherals on 128 MHz VPR, GPIO - 1x 64 MHz Port, 1.7 – 3.6V, 11 GPIOs

Low Leakage Peripherals:

2x QDEC, 7x Timer, Global RTC, 2x WDT, NFC A-Tag, TEMP, I2S, COMP, 3x PWM, LPCOMP, 14-bit 8CH ADC, 5x TWI/SPI/UART, GPIO (2x 16 MHz Port (P0, P1), 20 GPIO's, 1.7-3.6V, 32 GPIO)

Antenna choice

– integrated pre-certified

Chip antenna

or external antenna support via

Trace Pad

Ultra-small footprint

(6.3 mm x 7.9 mm x 1.75 mm)

Extended

Industrial

Temperature Rating

(-40° to +105 °C)

Development choice:

Zephyr RTOS, Nordic nRF Connect SDK & Canvas

Bluetooth LE

: Peripheral/Central, 2 Mbps (high throughput), LE Coded (long range), AoA/AoD, Mesh

Firmware Over the Air

(FOTA)

via MCUboot and Zephyr

Hostless operation

– Multi Core MCU reduces BOM

Fully featured development kits

to jump start Bluetooth LE development

Specification Summary

Processor / SoC / Chipset

MPU
Applications Core (High Performance) Arm Cortex-M33 with DSP, FPU, TrustZone support. 1524KB non-volatile RRAM 256KB RAM L1 cache 128MHz clock Uses voltage and clock frequency scaling
VPU
Software defined peripheral Core (ultra-low power) RISC-V CPU (VPR) fast lightweight peripheral processor (FLPR) dedicated for software defined peripherals 16MHz clock

Bluetooth

Standards
Bluetooth Core 6.2 – Single mode
Bluetooth LE Features
Channel Sounding GATT client/server – Any adopted/custom services Central/Peripheral roles Bluetooth LE mesh 2M PHY LE Coded PHY LE Advertising Extensions LE secure connections Data packet length extensions LE privacy v1.2 LE ping DTM Firmware (Test Modes)
PHY
2405–2480 MHz IEEE 802.15.4-2006 radio transceiver, implementing IEEE 802.15.4-2006 compliant 250kbps, 2450MHz, O-QPSK PHY Channels 11-26. Channel 11 2405MHz and CH26 2480MHz. Clear channel assessment (CCA) Energy detection (ED) scan CRC generation
Proprietary Radio Modes
2402–2480 MHz Nordic proprietary 1Mbps and 2Mbps modes radio transceiver 1Mbps nRF proprietary mode (ideal transmitter) 2Mbps nRF proprietary mode (ideal transmitter) 4Mbps nRF proprietary mode (ideal transmitter)
Frequency Range
2.402 - 2.480 GHz for BLE (CH0 to CH39) 2.405 - 2.480 GHz for IEEE 802.15.4-2006 PHY (CH11 to CH26)
Max Transmit Power
+8 dBm - Conducted 453-00223 (Integrated antenna) (Exclude antenna gain) +8 dBm - Conducted 453-00224 (External antenna) (Test at MHF4 connector)
Min Transmit Power
-40dBm
Receive Sensitivity
BLE 1 Mbps (BER=1E-3): -95dBm typical BLE 2 Mbps: -92dBm typical BLE 125 kbps: -103dBm typical BLE 500 kbps: -98.5dBm typical IEEE 802.15.4-2006 250kbps: TBD dBm typical
Supported Data Rates
1 Mbps BLE (over-the-air) 2 Mbps BLE (over-the-air) 125 kbps BLE (over-the-air) 500 kbps BLE (over-the-air) 250 kbps IEEE 802.15.4 802.15.4-2006 (over-the-air)

Nordic proprietary 1Mbps, 2Mbps and 4Mbps modes (over-the-air)

Link Budget

102 dB @ BLE 1 Mbps (TX power used is 8dBm)

110 dB @ BLE 125 kbps (TX power used is 7dBm)

NFC

NFC-Type A Listen mode compliant

Based on NFC forum specification:

13.56 MHz, Data rate 106 kbps, NFC Type2 and Type 4 emulation

Modes of Operation:

Disable, Sense, Activated

Use Cases:

Touch-to-Pair with NFC, NFC enabled Out-of-Band Pairing

Security

Designed for PSA Certified Level 3 with Secure Boot, Secure Firmware Update, and Secure Storage. Integrated tamper sensors detect attacks and take action, and cryptographic accelerators are hardened against side-channel attacks.

Radio Performance

Antenna Options

Internal - Chip antenna – on-board (453-00223 variant)

External - Connection via off module IPEX MHF4 (453-00224 RF trace pin variant)

Peripheral Interface
32x Multifunction I/O lines
GPIO Up to 32 multifunction GPIO's Applications Core (High Performance) 64MHz 1.7-3.6V GPIO port P2.00-P2.10 Software defined peripheral Core (ultra-low power) 16MHz 1.7-3.6V GPIO port P1.00-P1.15; P0.00-P0.04
ADC Applications Core (High Performance) 14-bit 20KS/s with oversampling 12-bit 250KS/s 10-bit 2MS/s AIN0-AIN7 pins up to 8 programmable gain channels
Global RTC (GRTC) Software defined peripheral Core (ultra-low power) Implements full real time clock and calendar as shared system time. Can run in System OFF mode. Ultra low power, 1us resolution, 52bits wide, uses 16MHz clock, 32.76kHz when other power modes.
RTC Software defined peripheral Core (ultra-low power) 2x low power runs off LFCLK
High Speed SPI/UART Applications Core (High Performance) 1x
PWM Software defined peripheral Core (ultra-low power) 3x 4channel PWM
SPI/UART/TWI Software defined peripheral Core (ultra-low power) 4x
TEMP Software defined peripheral Core (ultra-low power)
I2S Software defined peripheral Core (ultra-low power) 1x I2S (Inter-IC sound interface)
PDM Software defined peripheral Core (ultra-low power)

1x PDM (Pulse code modulation interface) for digital microphones
TIMER
Software defined peripheral Core (ultra-low power) 7x Timer (32bit)
QDEC
Software defined peripheral Core (ultra-low power) 2x QDEC (Quadrature decoder)
COMP
Software defined peripheral Core (ultra-low power) 1x COMP (comparator)
LPCOMP
Software defined peripheral Core (ultra-low power) 1x LPCOMP (low power comparator)
WDT
Software defined peripheral Core (ultra-low power) 2x WDT (Watchdog timer)
Wakeup Pins
Software defined peripheral Core (ultra-low power) 21
NFC A-Tag
Software defined peripheral Core (ultra-low power) 1x
Ext. 32.768 KHz Crystal
Software defined peripheral Core (ultra-low power) Not needed for normal radio operation. Optionally, connect +/-20ppm accuracy crystal for more accurate protocol timing. Fit associated load capacitor for crystal or use nRF54L15 internal load capacitor, which is configurable as 4 pF to 18 pF in 0.5pF steps on pins XL1, XL2.

Power

Input Voltage
1.7V-3.6V
Power Modes
Active Modes Peak Current (for maximum Tx power +8dBm) – Radio only 29 mA peak Tx, 9mA avg TX@ 1.8V, LE1M 23 mA peak Tx, 14.5mA avg TX@ 1.8V, CW 16 mA peak Tx, 5.7mA avg TX@ 3.3V, LE1M 13 mA peak Tx, 8.3mA avg TX@ 3.3V, CW Active Modes Peak Current (for Tx power -40dBm) – Radio only 9 mA peak Tx, 2.6mA avg TX @ 1.8V, LE1M 6.5 mA peak Tx, 3.2mA avg TX @ 1.8V, CW 6 mA peak Tx, 2.1mA avg TX @ 3.3V, LE1M 5 mA peak Tx, 2.6mA avg TX @ 3.3V, CW Ultra-low Power Modes System ON Idle - 3 uA (System ON IDLE with GRTC (XOSC) and 256 KB RAM) System OFF - 0.6 uA

Mechanical

Dimensions
7.9mm x 6.3mm x 1.75mm Pad Pitch – 0.65 mm Pad Type – Two rows of pads (LGA - Land Grid Array).
Weight
< 1 gram

Software

Security
Designed for PSA Certified Level 3 with Secure Boot, Secure Firmware Update, and Secure Storage. Integrated tamper sensors detect attacks and take action, and cryptographic accelerators are hardened against side-channel attacks.
Programmability Options
Via SWD (JTAG) 2 wire interface Nordic nRFConnect SDK: Software/Support available from Nordic directly https://devzone.nordicsemi.com/ Zephyr RTOS: Software/Support available from https://www.zephyrproject.org/ Canvas SW Suite : Software/ Support available from https://www.ezurio.com/canvas/software-suite
Firmware Update
Via SWD (JTAG) 2 wire interface or UART

Environmental

Operating Temperature
-40 °C to +105 °C
Storage Temperature
-40 °C to +105 °C
Lead Free
Lead-free and RoHS Compliant

Certifications

Regulatory Compliance
FCC/ISED/EU/UKCA/RCM/MIC/KC. Bluetooth and 802.15.4 Certified
Bluetooth SIG
Full Bluetooth SIG Declaration ID

Development

Development Kit
Development kit per module SKU (453-00223-K1 and 453-00224-K1 respectively) Bluetooth Channel Sounding Kit (in partnership with Metirionic)
Development Tools
Nordic nRFConnect - Android and iOS applications UART Firmware Upgrade Xbit Tools and utilities

Warranty

Warranty Terms
One Year Warranty

Functional Descriptions

Clocks and Timers

Clocks

HFXO - 32MHz crystal oscillator and nRF54L15 internal load capacitor setting

The BL54L15μ module contains the 32 MHz crystal, but the load capacitors to create 32MHz crystal oscillator circuit are inside the nRF54L15 chipset. Customer MUST set the internal nRF54L15 capacitors from 4 pF to 17 pF in 0.25pF if needed or use default value without changing.

The 32 MHz crystal inside the BL54L15μ module is a high accuracy crystal (± 15 ppm at room temperature) that helps with radio operation and reducing power consumption in the active modes.

LFCLK – Low Frequency clock source

There are four possibilities for the low frequency clock (LFCLK) and options are:

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LFRC (32.768kHz RC oscillator):

The Internal 32.768 kHz RC oscillator (LFRC) is fully embedded in nRF54L15 (and does not require additional external components) with an accuracy ± 250 ppm (after calibration of LFRC at least every eight seconds using the HFXO as a reference oscillator).

LF XO (32.768kHz crystal oscillator):

For higher LFCLK accuracy (greater than ± 250 ppm accuracy is required), the low frequency crystal oscillator (LF XO) must be used. To use LF XO, a 32.768kHz crystal must be connected between the XL1 and XL2 pins and the load capacitance between each crystal terminal and ground. Optionally internal (to nRF54L15) capacitor of maximum 18pF in 0.5pF steps are provided on pins XL1 and XL2.

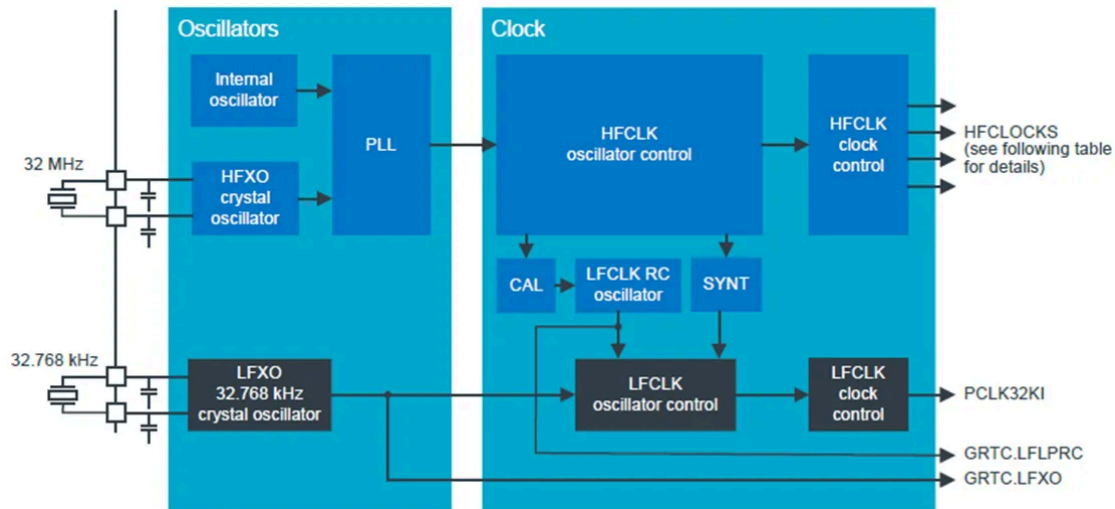
Low frequency (32.768 kHz) external source

: The 32.768 kHz oscillator (LF XO) is designed to work with external sources

LFSYNTH (32.768kHz Synthesised clock) from HFCLK (LFSYNTH):

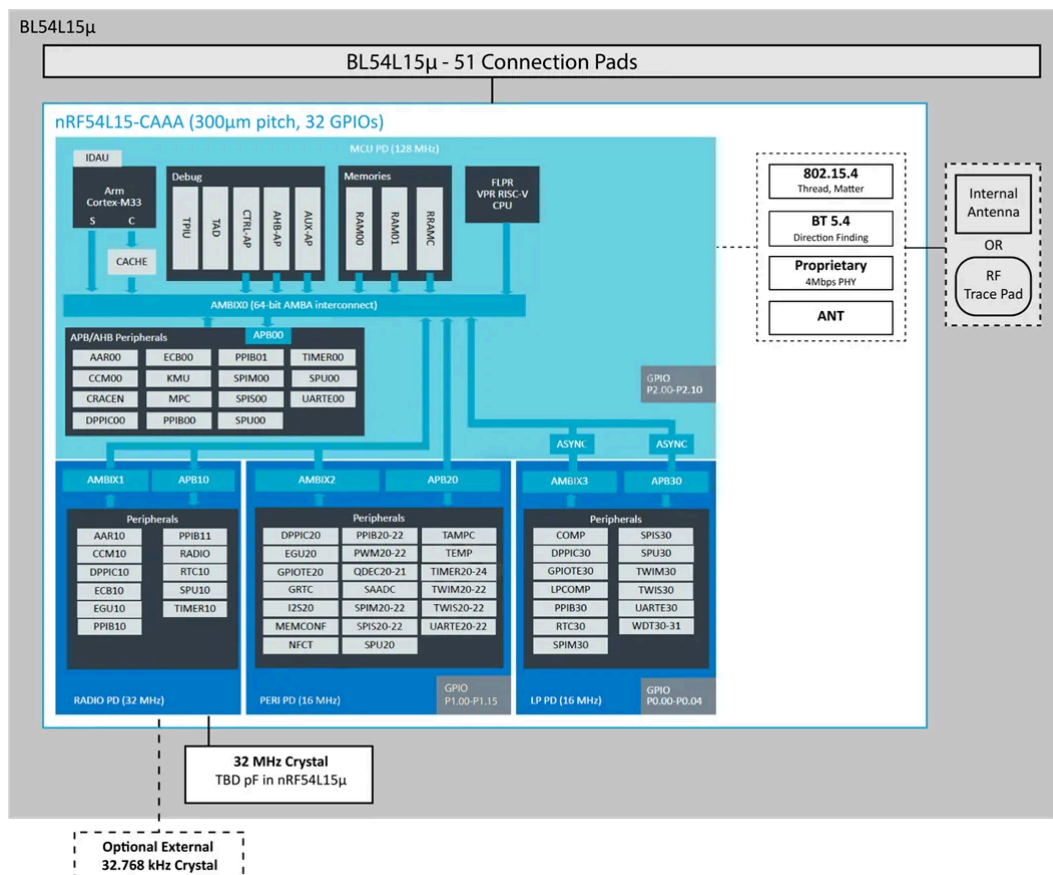
The LFCLK can be synthesised from the HFCLK source. LFSYNTH depends on the HFCLK to run. The accuracy of the LFCLK clock with LFSYNTH as a source assumes the accuracy of the HFCLK. If high accuracy is required, the HFCLK must generated from the HFXO. Using the LFSYNTH clock removes the requirement for an external 32.768kHz crystal but the increases the average power consumption as the HFCLK will be turned on in the system.

Other Internal Clocks



Hardware Architecture

Block Diagrams



Pin-Out

Pin #
Pin Name (bold pins for clock for interfaces, trace)
nRF54L15-CAAA 300um WLCSP, 32 GPIOs
nRF54L15-CAAA 300u pitch WLCSP, 32GPIOs Name
Description
Example usage
1
P2.05
G5
P2.05
General purpose I/O SPIM CS SPISCS UARTE RTS FLPR.5 QSPI CS
SPIM00/20 SPIS00/20 UARTE00/20 FLPR FLPR (QSPI)
2
P2.00
G6
P2.00
General purpose I/O SPIM DCX UARTE RXD FLPR.4 QSPI D3
SPIM00/20 UARTE00/20 FLPR FLPR (QSPI)
3
P1.08 /GRTCHFOUT/EXTREF
F7

P1.08
EXTREF
General purpose I/O
GRTC CLKOUTFAST
External reference for SAADC
Clock pin
4
GND
VSS
5
VDD_nRF
G3
VDD
1.7V-3.6V, connect external supply of 1.7V-3.6V to pin5(VDD_nRF)
6
P2.01
G7
P2.01
General purpose I/O
SPIM SCK
SPIS SCK
FLPR.0
QSPI SCK
Clock pin
SPIM00/20
SPIS00/20
FLPR
FLPR (QSPI)
7
P1.06/ASO[1]/AIN2
D7
P1.06
ASO[1]
AIN2
General purpose I/O
TAMPC active shield 1 output
Analog input

TAMPC
8
P1.07/ASI[1]/AIN3
E7
P1.07 ASI[1] AIN3
General purpose I/O TAMPC active shield 1 input Analog input
TAMPC
9
P1.03 /NFC2
D5
P1.03 NFC2
General purpose I/O NFC antenna connection
Clock pin
10
P1.01/XL2
C6
P1.01 XL2
General purpose I/O Connection for 32.768 kHz crystal
Ezurio Devkit: Optional 32.768kHz crystal pad XL2, XL1 and associated built-in configurable internal capacitors inside nRF54L15-CAAA chipset.
11
P1.02/NFC1
C7
P1.02 NFC1
General purpose I/O NFC antenna connection
12
P1.00/XL1

C5
P1.00
XL1
General purpose I/O Connection for 32.768 kHz crystal
Ezurio Devkit: Optional 32.768kHz crystal pad XL2, XL1 and associated built-in configurable internal capacitors inside nRF54L15-CAA chipset.
13
P2.02/SWO
F6
P2.02
General purpose I/O SPIM SDO SPIS SDO UARTE TXD FLPR.1 QSPI D0 Serial wire output (SWO)
SPIM00/20 SPIS00/20 UARTE00/20 FLPR FLPR (QSPI) Trace
14
GND
VSS
15
P1.05/ASI[0]/AIN1
E6
P1.05 ASI[0] RADIO[6] AIN1
General purpose I/O TAMPC active shield 0 input RADIO DFEGPIO Analog input
TAMPC RADIO

16
P1.15
B6
P1.15
General purpose I/O
17
P1.04 /ASO[0]/AIN0
D6
P1.04 ASO[0] AIN0
General purpose I/O TAMPC active shield 0 output Analog input
Clock pin TAMPC
18
P1.14/RADIO[5]/AIN7
B5
P1.14 RADIO[5] AIN7
General purpose I/O RADIO DFEGPIO Analog input
RADIO
19
P1.12 /ASI[3]/RADIO[3]/AIN5 Please see note 7
A3
P1.12 ASI[3] RADIO[3] AIN5
General purpose I/O TAMPC active shield 3 input RADIO DFEGPIO Analog input

Clock pin TAMPC RADIO
20
P1.13/RADIO[4]/AIN6
B4
P1.13 RADIO[4] AIN6
General purpose I/O RADIO DFEGPIO Analog input
RADIO
21
P1.10/ASI[2]/RADIO[1] Please see note 7
C3
P1.10 ASI[2] RADIO[1]
General purpose I/O TAMPC active shield 2 input RADIO DFEGPIO
TAMPC RADIO
22
P1.09/ASO[2]/RADIO[0] Please see note 7
B3
P1.09 ASO[2] RADIO[0]
General purpose I/O TAMPC active shield 2 output RADIO DFEGPIO
TAMPC RADIO
23
P1.11 /ASO[3]/RADIO[2]/AIN4 Please see note 7

C4
P1.11 ASO[3] RADIO[2] AIN4
General purpose I/O TAMPC active shield 3 output RADIO DFEGPIO Analog input
Clock pin TAMPC RADIO
24
GND
-
VSS
25
SWDCLK
E3
SWDCLK
Serial Wire Debug clock input with on-chip pull-up for debug and programming
26
P2.08/TRACEDATA[1]
D4
P2.08 TRACEDATA[1]
General purpose I/O FLPR.8 Trace data SPIM SDO SPIS SDO UARTE TXD
FLPR Trace SPIM00/21 SPIS00/21 UARTE00/21
27

SWDIO
F2
SWDIO
Serial Wire Debug IO with standard-drive and on-chip pull-down for debug and programming
28
GND
-
VSS
29
P0.04 /GRTCLFCLKOUT
D3
P0.04
General purpose I/O GRTC CLKOUT32K
Clock pin GRTC
30
P0.02
E2
P0.02
General purpose I/O
31
GND
-
VSS
32
nRESET
D2
nRESET
Pin RESET with internal pull-up resistor (13k Ohms). System Reset (Active Low).
33
P0.03 /GRTCPWM

E1
P0.03
General purpose I/O GRTC PWM
Clock pin GRTC
34
GND
-
VSS
35
P0.01
F1
P0.01
General purpose I/O
36
GND
-
VSS
37
RF_pad or NC
D1
ANT
Single ended radio antenna connection
RF pad active on BL54L15μ RF pin variant 453-00224. NC on BL54L15μ Integrated antenna variant 453- 00223.
38
P0.00
G1
P0.00
General purpose I/O
39
GND

-
VSS
40
P2.09/TRACEDATA[2]
F3
P2.09 TRACEDATA[2]
General purpose I/O FLPR.9 Trace data SPIM SDI SPIS SDI UARTE CTS
FLPR Trace SPIM00/21 SPIS00/21 UARTE00/21
41
P2.10/TRACEDATA[3]
G2
P2.10 TRACEDATA[3]
General purpose I/O FLPR.10 Trace data SPIM CS SPIS CS UARTE RTS
FLPR Trace SPIM00/21 SPIS00/21 UARTE00/21
42
P2.07/TRACEDATA[0]/SWO
E4
P2.07 TRACEDATA[0] SWO

General purpose I/O FLPR.7 Trace data Serial wire output (SWO) SPIM DCX UARTE RXD
FLPR Trace Trace SPIM00/21 UARTE00/21
43
P2.06/TRACECLK
F4
P2.06 TRACECLK
General purpose I/O FLPR.6 SPIM SCK SPIS SCK Trace clock
Clock pin FLPR SPIM00/21 SPIS00/21 Trace
44
P2.03
E5
P2.03
General purpose I/O FLPR.3 QSPI D2
FLPR FLPR (QSPI)
45
P2.04
F5
P2.04
General purpose I/O SPIM SDI SPIS SDI

UARTE CTS
FLPR.2
QSPID1
SPIM00/20
SPIS00/20
UARTE00/20
FLPR
FLPR (QSPI)
G1
GND
-
VSS
G2
GND
-
VSS
G3
GND
-
VSS
G4
GND
-
VSS
G5
GND
-
VSS
G6
GND
-
VSS

Notes:

Note 1 - GPIO

GPIO = General Input or Output (GPIO level voltage tracks VDD pin). AIN = Analog input.

If GPIO is selected as an input, ensure the input is not floating (which can cause current consumption to drive with time in low power modes (such as System ON Idle), by selecting the internal pull up or pull down.

Must connect all GND pads to host board PCB GND plane.

Note 2 - Clock for serial interfaces or trace

Some peripherals (SPI, TWI, PDM, I2S, TRACE, GRTC) have clock signals. Dedicated clock pins have been optimized to ensure correct timing relationship between clock and data signal for these peripherals. Pins that can be used as clock signals are shown with pin name in red colour.

The peripheral data signal must be configured to use pins close to the clock pin. This ensures that the internal paths from the peripheral to the pin have the same delay, so that the data and clock signals reach the pins at the same time.

For high-speed signals, the printed circuit board (PCB) layout must use short PCB traces of identical length. This makes sure any delays are kept to a minimum and it assures close to identical delay and clock path.

Note 3 - Dedicated pins

UARTE20/21: Can use any pin on GPIO port P1. Can be connected across power domains to dedicated pin on P2.

SPIM00: Has dedicated pins on GPIO port P2. For 32MHz operation, the pins must be configured using extra high drive E0/E1 configuration.

SPIM20/21: Can use any pin on GPIO port P1. Can be connected across power domains to dedicated pin on P2.

SPIS20/21: Can use any pin on GPIO port P1. Can be connected across power domains to dedicated pin on P2.

TRACE: Has dedicated pins on GPIO port P2. For 32MHz operation, the pins must be configured using extra high drive E0/E1 configuration.

GRTC: Has dedicated pins for clock and PWM output.

TAMPC: Has dedicated pins for active shield inputs and outputs.

FLPR: Uses dedicated pins on GPIO port P2 for emulated peripherals such as QSPI.

RADIO: Uses dedicated pins on GPIO port P1 for antenna switch control (DFEGPIO for direction finding).

NFC: Uses dedicated pin listed in pin definitions table.

Note 4 - SWDIO / SWCLK / nRESET / VDD / GND

Customer MUST bring out SWDIO, SWCLK, nRESET, VDD, GND for programming purposes.

Note 5 - GPIO P2.15

P1.15 GPIO is only available on the BL54L15μ which uses nRF54-CAAA(300um pitch) which as 32 GPIO's.

Note 6 - RF_PAD

RF_pad (pin37) is for the BL54L15μ RF pad variant (453-00224) module only. If using the BL54L15μ module RF pad variant (453-00224), customer MUST copy the 50-Ohms GCPW RF track design, detailed in 50-Ohms RF Trace on Host PCB for BL54L15μ RF pad variant (453-00224)

Note 7

If GPIOs P1.09, P1.10, P1.11, and P1.12 are used, please apply the following operations.

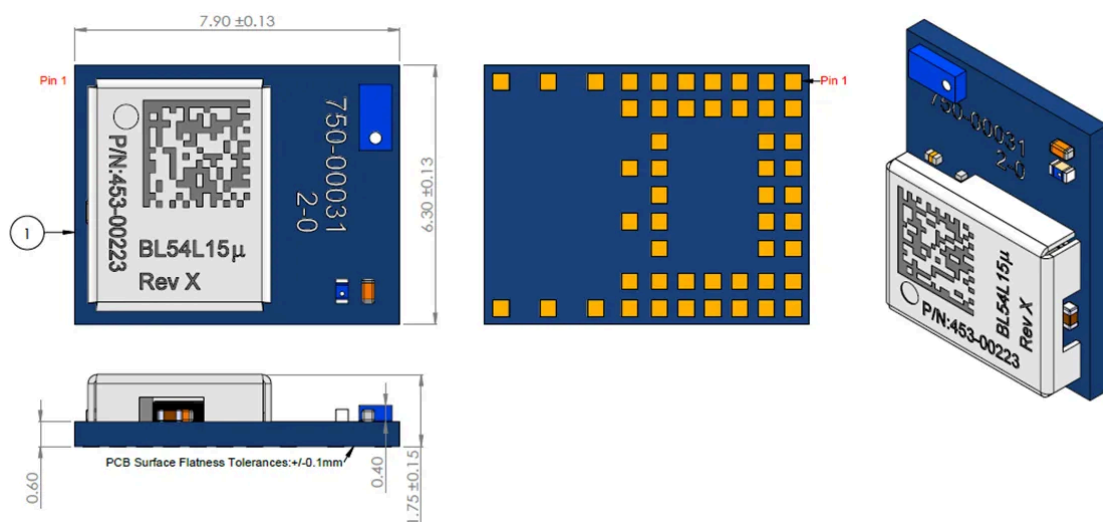
Add a 330 Ω or larger series resistance directly after the tracks enter the outer layer of the PCB.

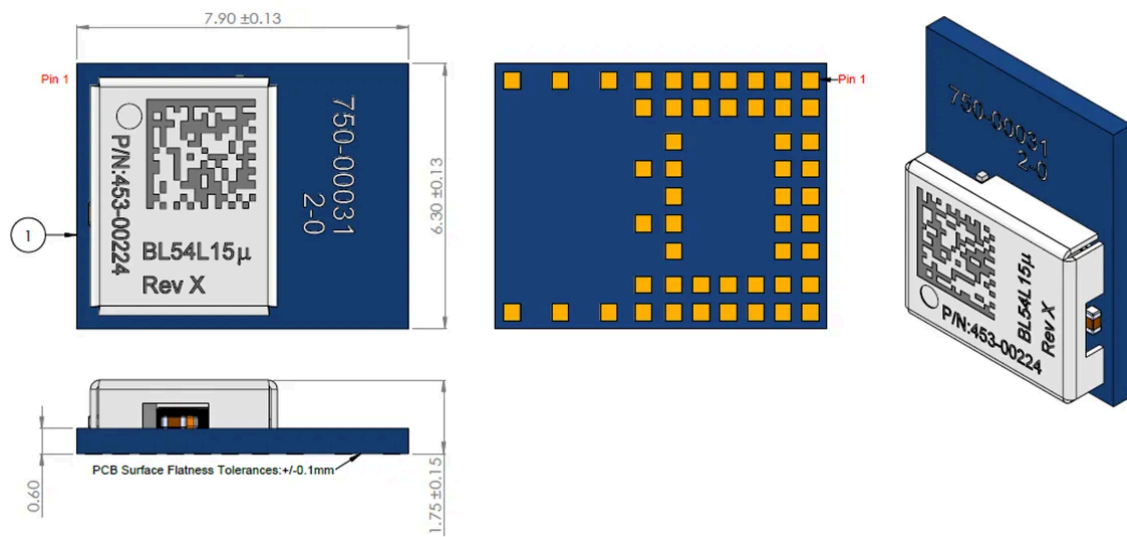
Keep PCB tracks as short as possible and in between ground layers if possible.

Use toggling speed lower than 1 MHz

If unused, leave these pins unconnected.

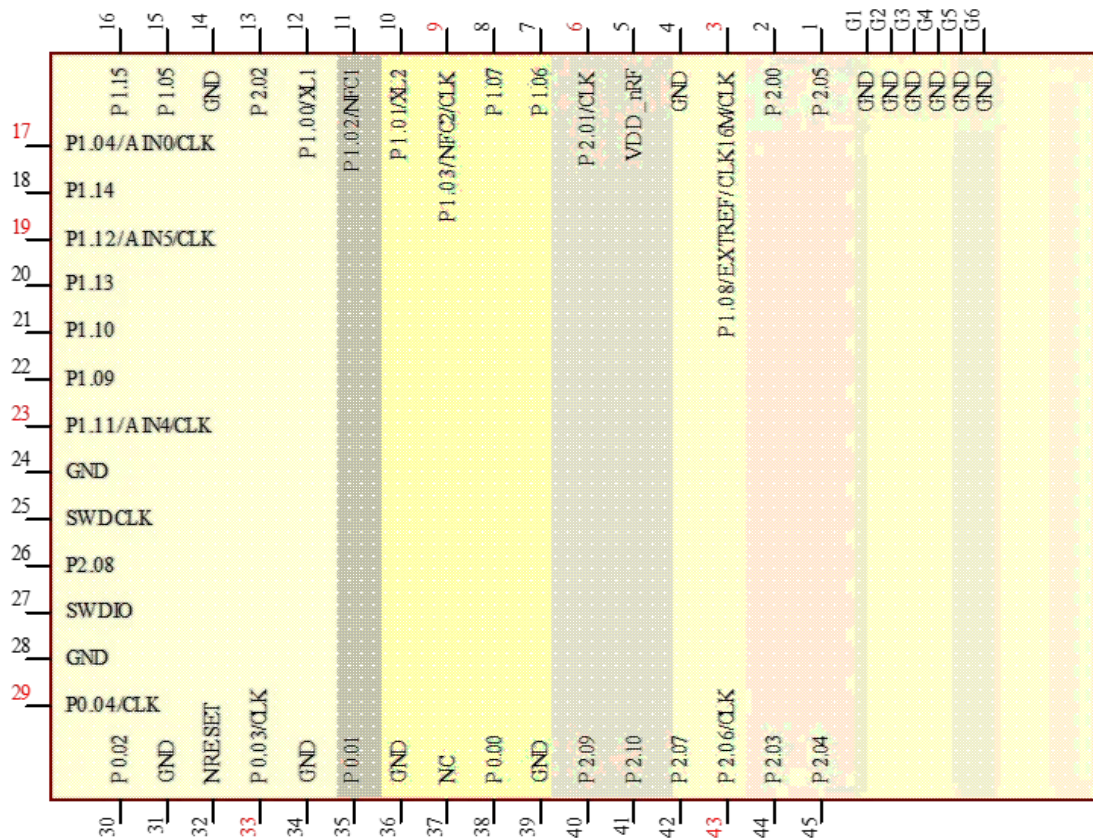
Mechanical Drawings

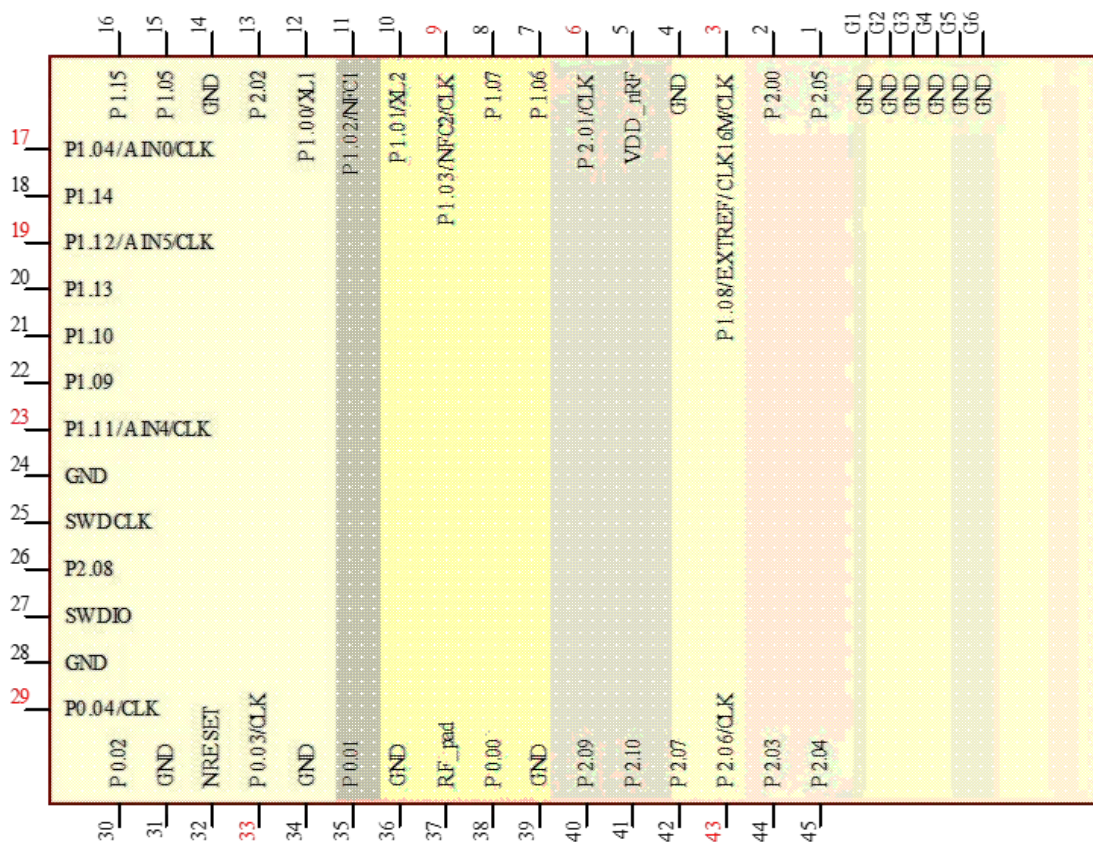




3D models for
 BL54L15μ Module, RF Trace Pin (453-00224)
 and
 BL54L15μ Module, Chip Antenna (453-00223)
 on the BL54L15μ product page –
<https://www.ezurio.com/product/BL54L15μ-series-bluetooth-le-80215-4-nfc>

Schematics





Electrical Characteristics

Absolute Maximum Ratings

Absolute maximum ratings are the extreme limits for supply voltage and voltages on digital and analogue pins of the module are listed below; exceeding these values causes permanent damage.

Parameter
Min
Max
Unit
Supply Voltages
VDD_nRF
-0.3
3.9
V
VDD_nRF ext
-0.3
3.7
V
I/O pin voltage
Voltage at GPIO pin (at VDD≤3.6V)
-0.3
VDD_nRF + 0.3
V
Voltage at GPIO pin (at VDD > 3.6V)
3.9
V
Voltage at GPIO pin under extended operating temperature (at VDD≤3.4V)
-0.3
VDD_nRF + 0.3
V
Voltage at GPIO pin under extended operating temperature (at VDD > 3.4V)
-0.3
3.7
V
NFC antenna pin current (NFC1/2)

-
130
mA
Environmental
Storage temperature
-40
+125
°C
MSL (Moisture Sensitivity Level)
-
4
-
ESD (as per EN301-489)
Conductive
Air Coupling
4
8
kV
kV
Flash Memory (Endurance) (Note 2)
10,000
Write/erase cycles
Flash Memory (Retention)
10 years at 85°C
2 years at 105°C
years at TBD °C

Notes:

The absolute maximum rating for VDD pin (max) is 3.9 V for the BL54L15μ.

Wear levelling can be implemented by customer.

Recommended Operating Conditions

Parameter
Min
Typ
Max
Unit
VDD_nRF (independent of DCDC) supply range
1.7
3.6
V
VDD_nRF ext (VDD_nRF supply voltage under extended operating temperature)
1.7
3.4
V
VDD_nRF Maximum ripple or noise (See Note 1)
-
-
10
mV
Operating Temperature Range
-40
+25
+85
°C
Extended operating temperature
85
105
°C

 Notes:

- This is the maximum VDD_nRF ripple or noise (at any frequency) that does not disturb the radio.
- The on-board power-on reset circuitry may not function properly for rise times longer than the specified maximum.
- BL54L15μ power supply: Connect external supply voltage (within range 1.7V-3.6V) to VDD_nRF pin. Please note an additional parameter for the extended operating temperature condition is provided.

Chat

For the best on-board chip antenna performance, the module 453-00223 MUST be placed on the edge of the host PCB and preferably in the edge centre and host PCB.

Note 3:

BL54L15µ development board has the 453-00223 placed on the edge of the PCB board (and not in corner).

Note 4:

Ensure that there is no exposed copper under the module on the host PCB.

Note 5:

You may modify the PCB land pattern dimensions based on their experience and/or process capability.

Mandatory SW requirements related to hardware

32MHz crystal internal load capacitor setting

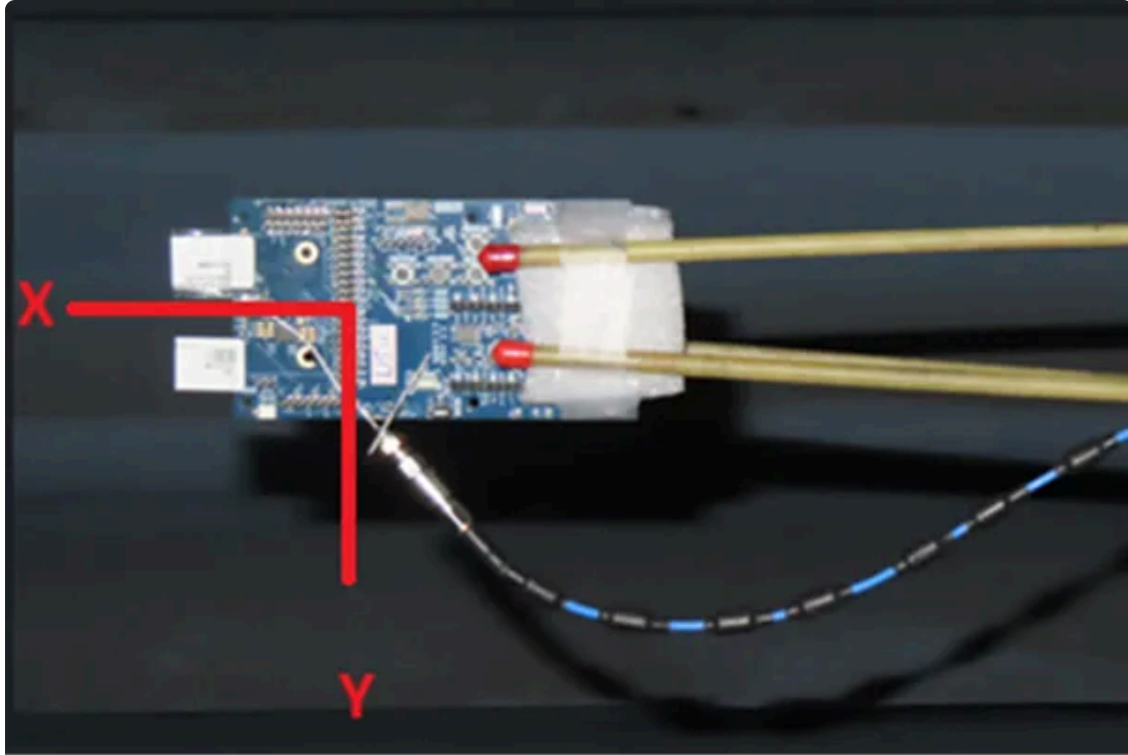
MANDATORY. BL54L15µ module contains the 32 MHz crystal but the load capacitors to create 32 MHz crystal oscillator circuit are inside the nRF54L15-CAAA chipset. Customer MUST set the internal nRF54L15 capacitors to from 4 pF to 17 pF in 0.25pF if needed, or use default value without changing.

Antenna Characteristics

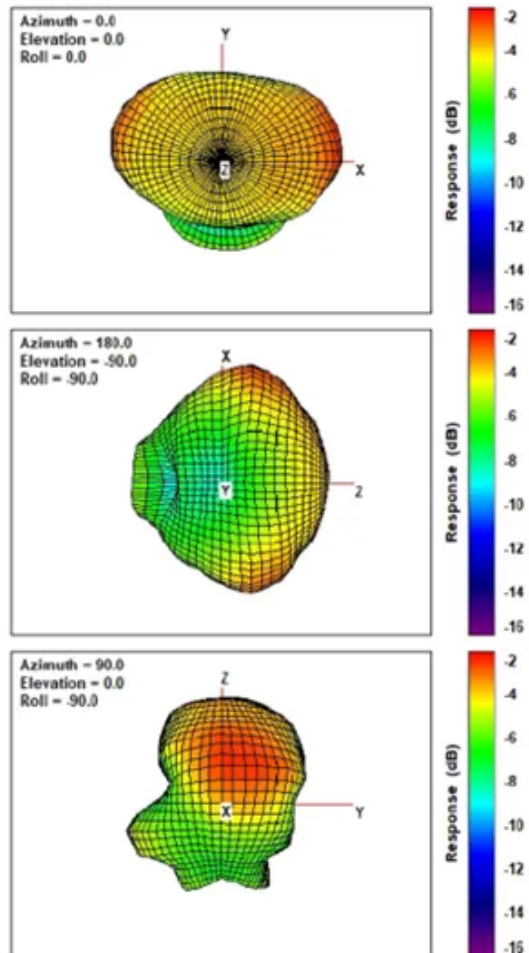
Summary of Antenna Performance

2402MHz Peak
2402MHz Avg
2440MHz Peak
2440MHz Avg
2480MHz Peak
2480MHz Avg
453-00223 Chip antenna
-1.82dBi
-5.1dBi
-2.05dBi
-5.52dBi
-2.51dBi
-6.46dBi

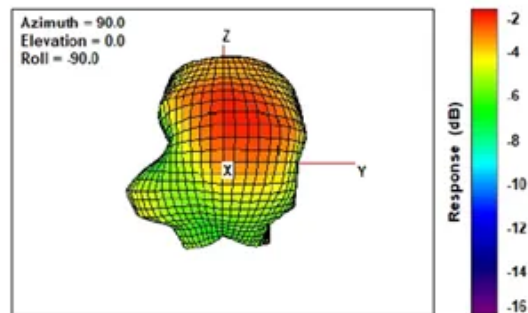
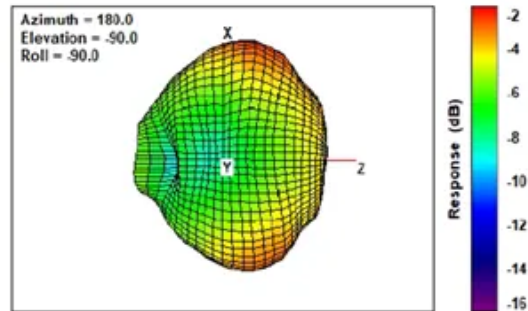
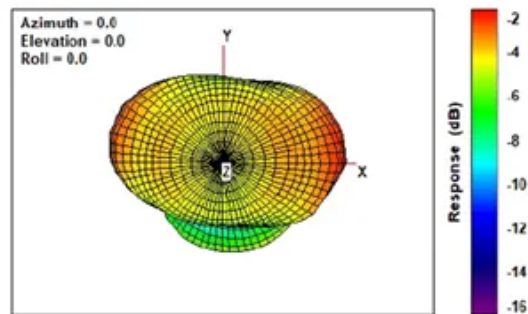
2.4GHz Radiated Performance



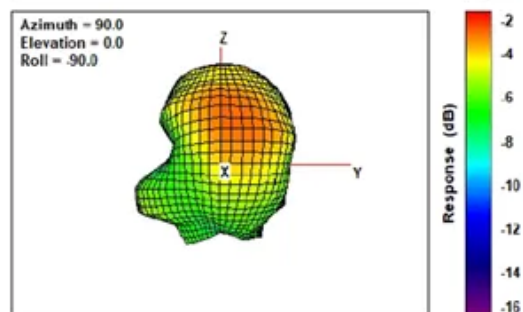
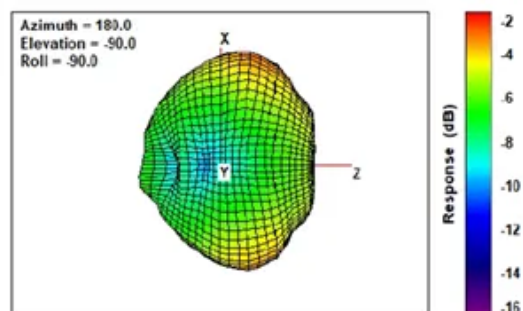
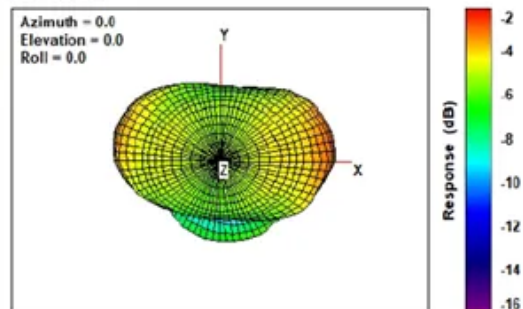
2400MHz

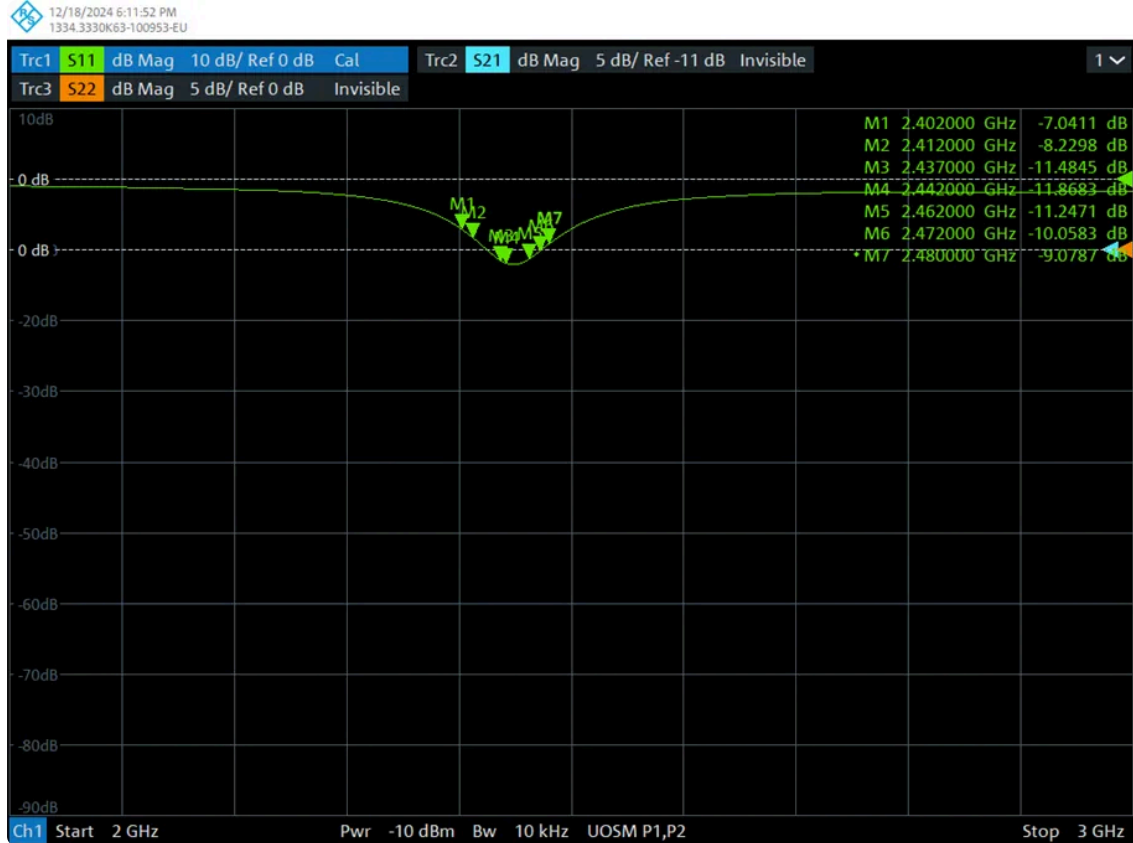


2450MHz



2500MHz





Circuit (Overview and Checklist)

The BL54L15 μ is easy to integrate, requiring one mandatory external 10 μ F capacitor on customers board and apart from that those components which customer require for development and in your end application.

The following are suggestions for your design for the best performance and functionality.

BL54L15 μ power supply:

External supply voltage (1.7V-3.6V) is connected to VDD_nRF pin(pin5).

External power source should be within the operating range, rise time and noise/ripple specification of the BL54L15 μ . Add decoupling capacitors for filtering the external source. Power-on reset circuitry within BL54L15 μ series module incorporates brown-out detector, thus simplifying your power supply design. Upon application of power, the internal power-on reset ensures that the module starts correctly.

AIN (ADC) and GPIO pin IO voltage levels

BL54L15 μ GPIO voltage levels are at VDD. Ensure input voltage levels into GPIO pins are at VDD also (if VDD source is a battery whose voltage drops). Ensure ADC pin maximum input voltage for damage is not violated.

AIN (ADC) impedance and external voltage divider setup

If you need to measure with ADC a voltage higher than 3.6V, you can connect a high impedance voltage divider to lower the voltage to the ADC input pin.

SWD

This is REQUIRED for loading firmware. MUST wire out the SWD two wire interface on host design. Five lines should be wired out, namely SWDIO, SWDCLK, nRESET, GND and VDD.

UART and flow control (CTS, RTS)

Required if customer requires UART.

TWI (I2C)

It is essential to remember that pull-up resistors on both SCL and SDA lines are required, the value as per I2C standard. BL54L15 μ (nRF54L15-CAAA) can provide 13K Ohms typical pull up values internally. For other values, fit external pull-up resistor on both SCL and SDA as per I2C specification to set speed. The I2C specification allows a line capacitance of 400pF.

QSPI, High Speed SPI, High speed TWI (I2C, 1Mbps) and Trace

High-Speed SPI, TWI and Trace come on dedicated GPIO pins only. Other lower speed SPI and TWI can come out on any GPIO pins. For all high-speed signal, the printed circuit board (PCB) layout must ensure that connections are made using short PCB traces.

GPIO pins

If GPIO is selected as an input, ensure the input is not floating (which can cause current consumption to drive with time in low power modes (such as System ON Idle), by selecting the internal pull up or pull down.

NFC antenna connector

To make use of the Ezurio flexi-PCB NFC antenna (part # 0600-00061), fit connector:

Description – FFC/FPC Connector, Right Angle, SMD/90d, Dual Contact, 1.2 mm Mated Height

Manufacturer – Molex

Manufacturers Part number – 512810594

Add tuning capacitors of 300 pF on NFC1 pin to GND and 300 pF on NFC2 pins to GND if the PCB track length is similar as development board.

nRESET pin (active low)

Hardware reset. Wire out to push button or drive by host.

By default module is out of reset when power applied to VDD pins (13K pull-up inside BL54L15μ (nRF54L15-CAAA)).

Optional External 32.768kHz crystal

If the optional external 32.768kHz crystal is needed, then use a crystal that meets specification and add load capacitors (either inside nRF54L15-CAAA or discrete capacitors outside BL54L15μ (nRF54L15-CAAA) whose values should be tuned to meet all specification for frequency and oscillation margin.

BL54L15μ module RF pad variant (453-00224)

RF_pad (pin37) is for the BL54L15μ RF pad variant (453-00224) module only. If using the BL54L15μ module RF pad variant (453-00224), customer MUST use 50-Ohms RF track design, detailed in the following section: 50-Ohms RF Trace on Host PCB for BL54L15μ RF pad variant (453-00224)

PCB Layout

PCB Layout on Host PCB - General

Checklist (for PCB):

MUST locate BL54L15μ module close to the edge of PCB (mandatory for the 453-00223 for on-board chip antenna to radiate properly).

Use solid GND plane on inner layer (for best EMC and RF performance).

All module GND pins MUST be connected to host PCB GND.

Place GND vias close to module GND pads as possible.

Unused PCB area on surface layer can be flooded with copper but place GND vias regularly to connect the copper flood to the inner GND plane. If GND flood copper is on the bottom of the module, then connect it with GND vias to the inner GND plane.

Route traces to avoid noise being picked up on VDD supply and AIN (analogue), GPIO (digital) traces and high-speed traces.

Ensure no exposed copper is on the underside of the module (refer to land pattern of BL54L15μ development board).

PCB Layout on Host PCB - 453-00223

Antenna Keep-Out on Host PCB

The 453-00223 has an integrated chip antenna and its performance is sensitive to host PCB. It is critical to locate the 453-00223 on the edge of the host PCB (or corner) to allow the antenna to radiate properly. Refer to guidelines in section

[Host PCB Land Pattern and Antenna Keep-out for the 453-00223](#)

. Some of those guidelines repeated below.

Ensure there is no copper in the antenna keep-out area on any layers of the host PCB. Keep all mounting hardware and metal clear of the area to allow proper antenna radiation.

For best antenna performance, place the 453-00223 module on the edge of the host PCB, preferably in the edge center.

The BL54L15μ development board (453-00223-K1) has the 453-00223 module on the edge of the board (not in the corner). The antenna keep-out area is defined by the BL54L15μ development board which was used for module development and antenna performance evaluation shown below, where the antenna keep-out area is 3mm wide, 5mm long; with PCB dielectric (no copper) height ~1.6mm sitting under the 453-00223 chip antenna module.

The 453-00223 chip antenna is tuned when the 453-00223 is sitting on development board (host PCB) with size of 113mm x 63.5mm x 1.6mm.

A different host PCB thickness dielectric will have small effect on antenna.

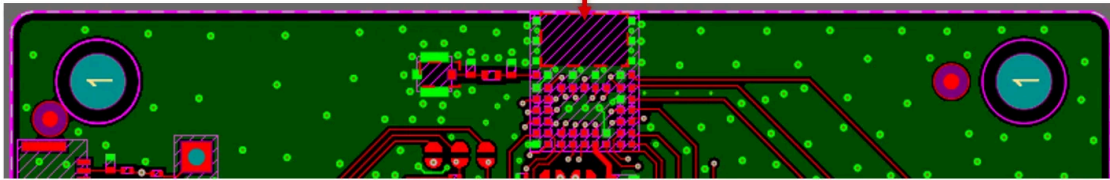
The antenna-keep-out defined in the

[Host PCB Land Pattern and Antenna Keep-out for the 453-00223 section](#)

.

Host PCB land pattern and antenna keep-out for the BL54L15μ applies when the 453-00223 is placed in the edge of the host PCB preferably in the edge center.

Antenna Keep-out



Notes:

The BL54L15μ module is placed on the edge, preferably edge centre of the host PCB.

Copper cut-away on all layers in the Antenna Keep-out area under the 453-00223 on host PCB.

Antenna Keep-out and Proximity to Metal or Plastic

Checklist (for metal /plastic enclosure):

Minimum safe distance for metals without seriously compromising the antenna (tuning) is 40 mm top/bottom and 30 mm left or right.

Metal close to the 453-00223 Chip antenna (bottom, top, left, right, any direction) will have degradation on the antenna performance. The amount of that degradation is entirely system dependent, meaning you will need to perform some testing with your host application.

Any metal closer than 20 mm will begin to significantly degrade performance (S11, gain, radiation efficiency).

It is best that you test the range with a mock-up (or actual prototype) of the product to assess effects of enclosure height (and materials, whether metal or plastic) and host PCB ground (GND plane size).

External Antenna Integration

50-Ohms RF Trace on Host PCB for BL54L15μ RF pad variant (453-00224)

To use an external antenna requires BL54L15μ module variant with RF trace pad (453-00224) and 50-Ohm RF trace from RF_pad (pin37) of the module (BL54L15μ 453-00054) to RF antenna connector (IPEX MHF4) on host PCB. On this RF path, BL54L15μ module GND pin36, pin39 and pin G1 used to support 50-Ohm RF trace.

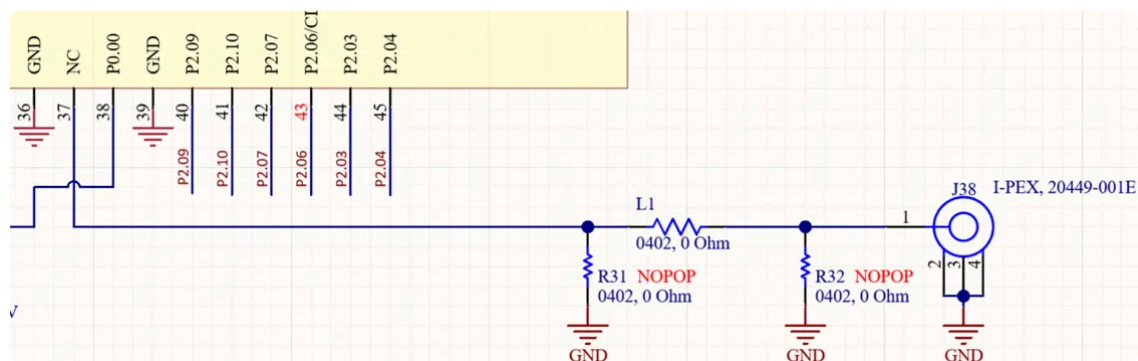
Checklist for SCH

Suggest fit RF connector IPEX MHF4 Receptacle (MPN: 20449-001E),

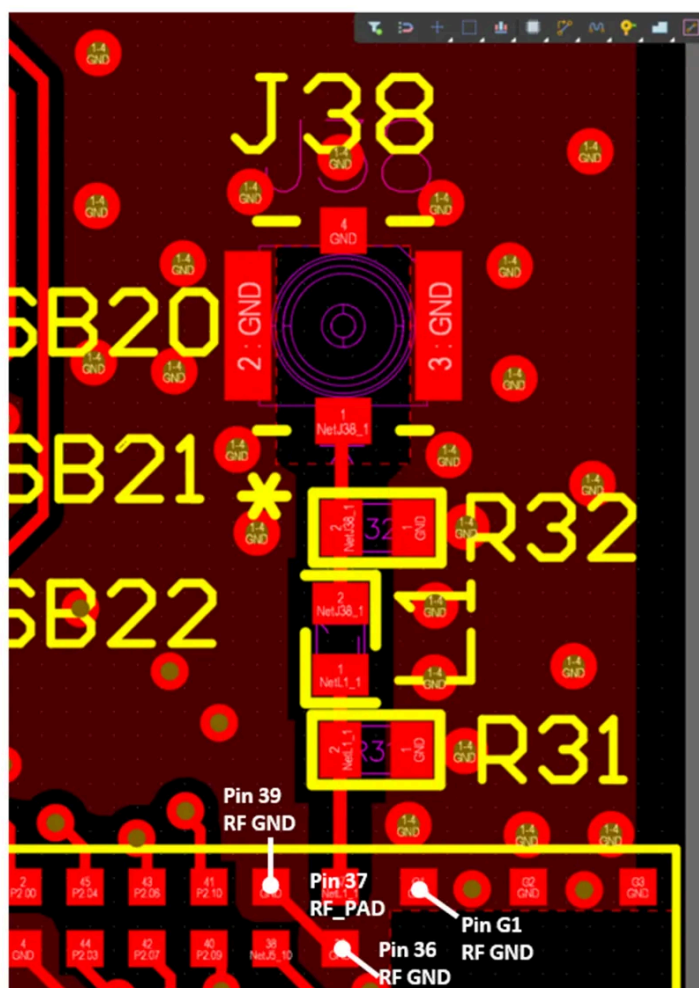
<https://www.i-pex.com/product/mhf-4-smt#>

!

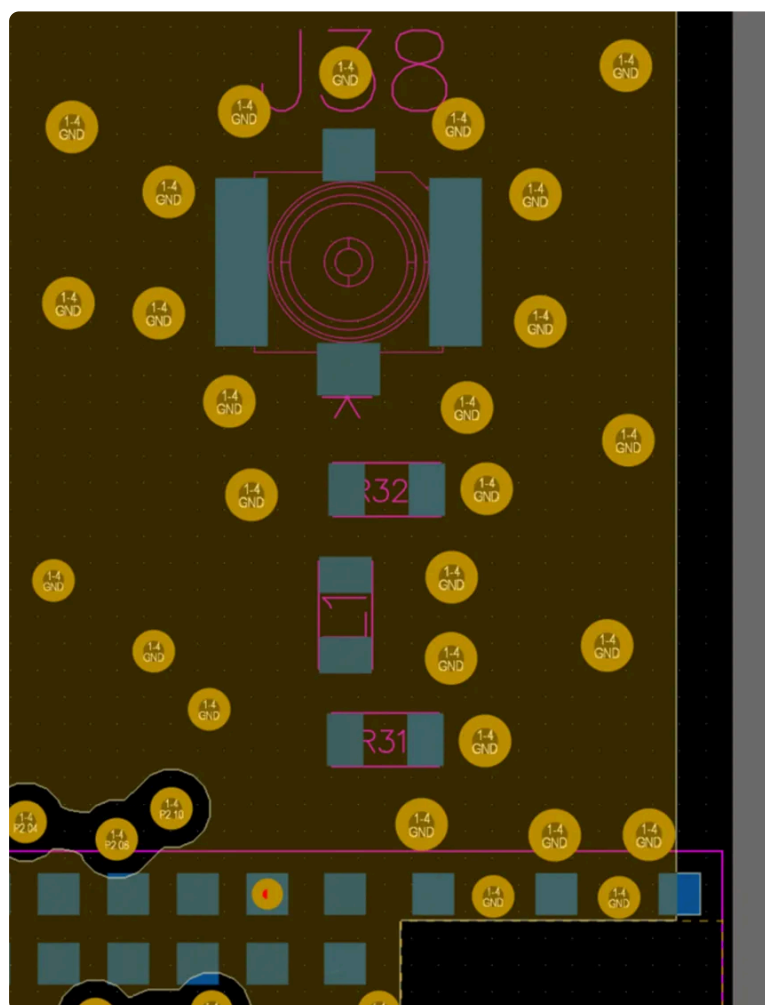
Suggest preserve pi network for tuning as shown below



Layer1 (RF Track and RF GND)

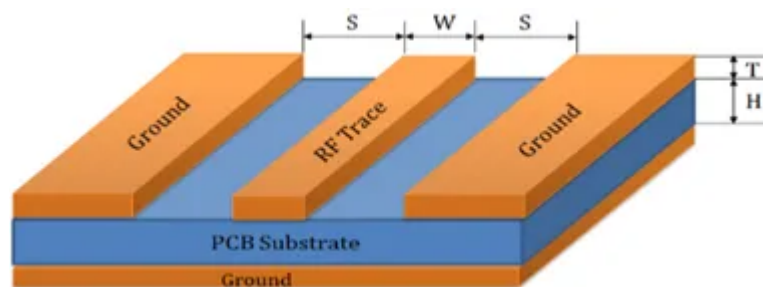


Layer2 (RF GND) and Layer2 copper cut-out under RF connector



MUST use a 50-Ohm RF trace (GCPW, that is Grounded Coplanar Waveguide) from RF_pad (pin37) of the module (BL54L15μ 453-00224) to RF antenna connector (IPEX MHF4) on host PCB.

- To ensure regulatory compliance, MUST follow exactly the following considerations for 50-Ohms RF trace design and test verification:



Propose PCB Stack Up					
Layer	Type	Thickness (mil)			Dielectric
Top side solder mask		0.8	mils		4.4
L1		1/3oz copper+ plating	1.2	mils	
		NPG151 PP 3313	4	mils	4.08
L2		copper	1.4	mils	
		NPG151 CORE 1.3mm	48.38	mils	4.47
L3		copper	1.4	mils	
		NPG151 PP 3313	4	mils	4.08
L4		1/3oz copper+ plating	1.2	mils	
Bottom side solder mask		0.8	mils		4.4
TOTAL		63.18	mils		
		1.60	mm		

Note 1:

The plating (ENIG) above base 0.5 oz copper is not listed, but plating expected to be ENIG.

The 50-Ohms RF trace design MUST be Grounded Coplanar Waveguide (GCPW) with:

Layer1 RF track width (W) of 6.0 mil and

Layer1 gap (G) to GND of 10 mil and where the

Layer1 to Layer 2 dielectric thickness (H) MUST be 4 mil (dielectric constant Er 4.08).

Further the Layer1 base copper must be 0.33-ounce base copper (that is 0.46 mil) plus the plating

Layer1 MUST be covered by solder mask of 0.8 mil thickness (dielectric constant Er 4.4).

The 50-Ohms RF trace design MUST follow the PCB stack-up shown above.

(Layer1 to Layer2 thickness MUST be identical to the BL5415u development board).

The 50-Ohms RF track should be a controlled-impedance trace e.g., ±10%.

Place GND vias regularly spaced either side of 50-Ohms RF trace to form GCPW (Grounded coplanar waveguide) transmission line as shown above and use BL54L15u module GND pin 36, pin 39 and pin G1.

Use spectrum analyzer to confirm the radiated (and conducted) signal is within the certification limit.

External Antenna Integration with BL54L15μ RF trace pin variant (453-00224)

Please refer to the regulatory sections for FCC, ISDE, CE, MIC, UKCA and RCM details of use of BL54L15μ with external antennas in each regulatory region.

The BL54L15μ family has been designed to operate with the below external antennas (with a maximum gain of 2.32 dBi). The required antenna impedance is 50 ohms. See the table below. External antennas improve radiation efficiency.

Manufacturer
Model
Ezurio Part Number
Type
Connector
Peak Gain @ 2400-2500 MHz
Peak Gain @ 2400-2480 MHz
Ezurio (Laird Connectivity)
NanoBlue
EBL2400A1-10MH4L
PCB Dipole
IPEX MHF4
2 dBi
-
Ezurio (Laird Connectivity)
FlexPIFA
001-0022
PIFA
IPEX MHF4
-
2 dBi
Mag.Layers
EDA-8709-2G4C1-B27-CY
0600-00057
Dipole
IPEX MHF4
2.32 dBi
-
Ezurio (Laird Connectivity)
mFlexPIFA
EFA2400A3S-10MH4L
PIFA
IPEX MHF4
-
2 dBi
Ezurio (Laird Connectivity)

iFlexPIFA Mini
EFG2401A3S-10MH4L
PIFA
IPEX MHF4
-
2 dBi
Ezurio (Laird Connectivity)
Ezurio NFC
0600-00061
NFC
N/A
-
-

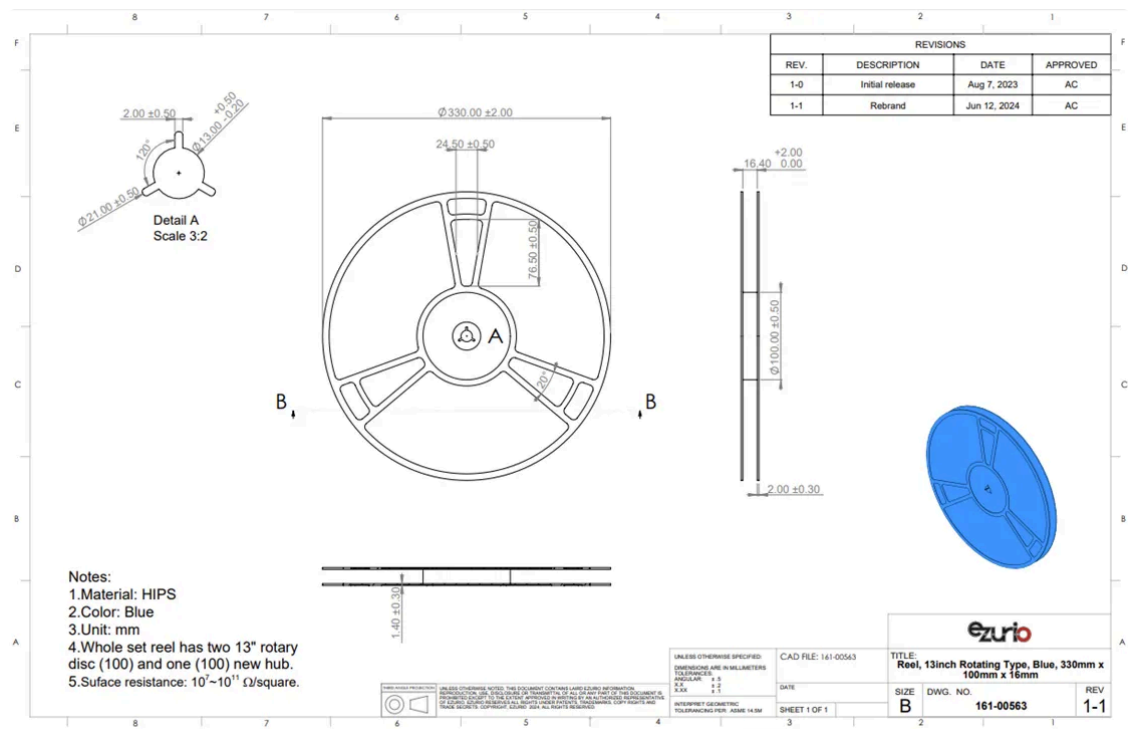
Application Note for Surface Mount Modules

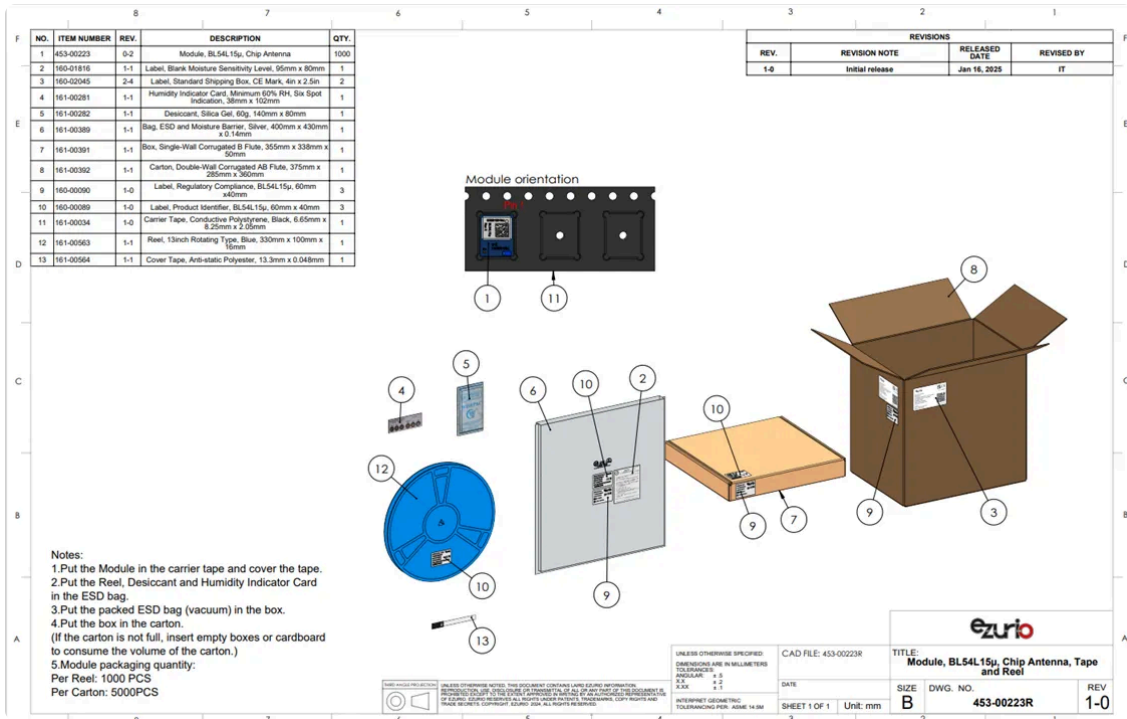
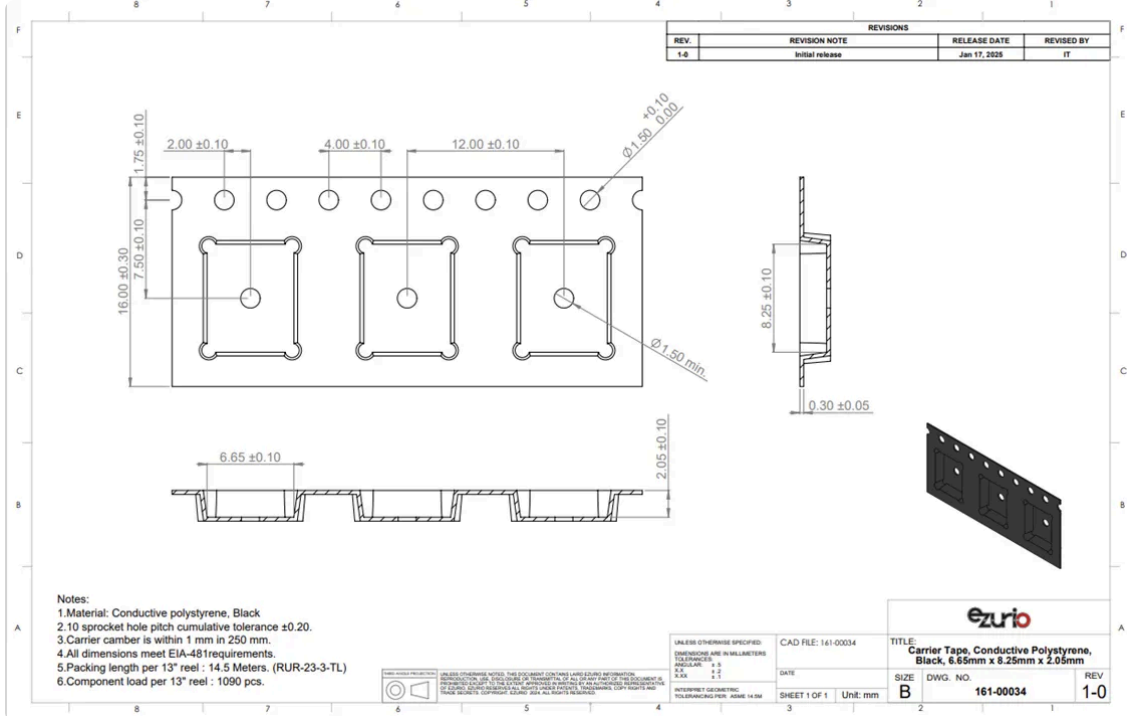
Introduction

Ezurio's surface mount modules are designed to conform to all major manufacturing guidelines. This application note is intended to provide additional guidance beyond the information that is presented in the user manual. This application note is considered a living document and will be updated as new information is presented.

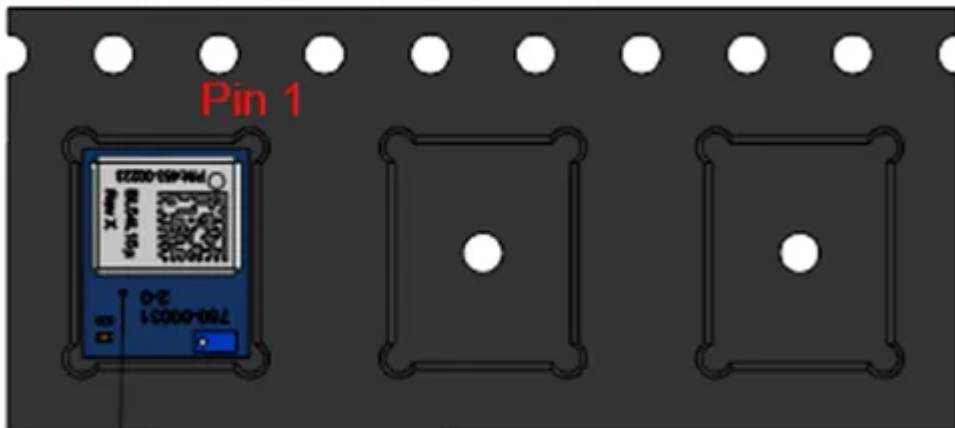
The modules are designed to meet the needs of several commercial and industrial applications. They are easy to manufacture and conform to current automated manufacturing processes.

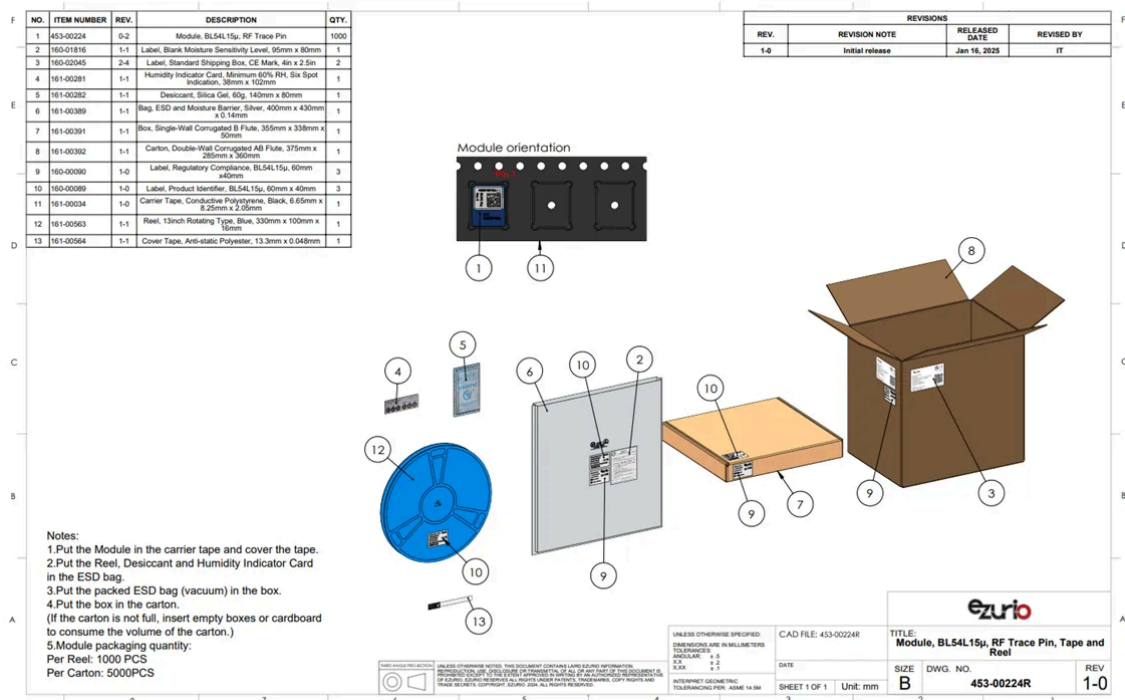
Part numbers – 453-00223R and 453-00224R are shipped as Tape / Reel, with a reel containing 1,000 pcs.



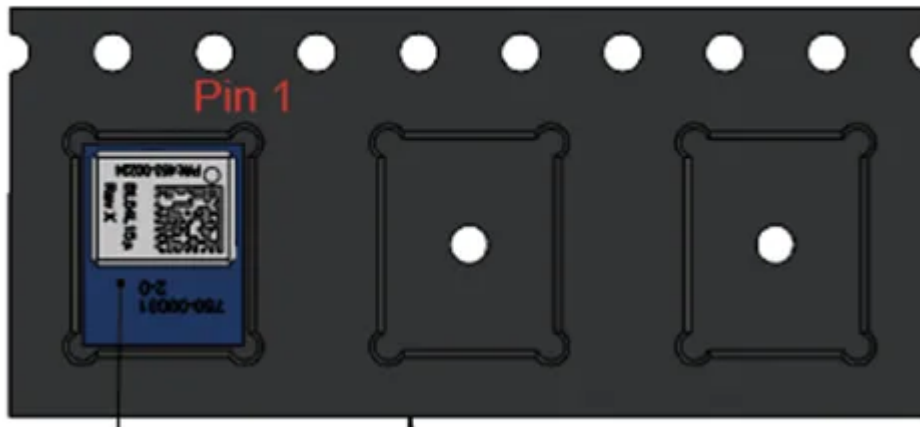


Module orientation





Module orientation



Shipping

All modules are shipped in tape and reel package and sealed in ESD Bags.

Labeling

MSL Label

The following labels are placed on the anti-static bag. The BL54L15u solder-down modules are classified as MSL4 devices.



Caution
This bag contains
MOISTURE-SENSITIVE DEVICES

LEVEL



If blank, see adjacent
bar code label

1. Calculated shelf life in sealed bag : 12 months at $<40^{\circ}\text{C}$ and $<90\%$ relative humidity (RH)
2. Peak package body temperature: _____ $^{\circ}\text{C}$
If blank, see adjacent bar code label
3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be
 - a) Mounted within: _____ hours of factory conditions
If blank, see adjacent bar code label
 $\leq 30^{\circ}\text{C}/60\% \text{ RH}$, or
 - b) Stored per J-STD-033
4. Devices require bake, before mounting, if:
 - a) Humidity Indicator Card reads $> 10\%$ for level 2a - 5a devices or $> 60\%$ for level 2 devices when read at $23 \pm 5^{\circ}\text{C}$
 - b) 3a or 3b are not met.
5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure.

Bag Seal Date: _____
If blank, see adjacent bar code label

Note: Level and body temperature defined by IPC/JEDEC J-STD-020

Product Identifier Labels

M/N:BL54L15u Rev 1

ezurio

P/N:453-00223R

D/C:SSYYWWD

CE **UK**
CA

Q'TY:XXXXPCS

BOX_ID:BXXXXXXYMDXXXXXXX



M/N:BL54L15u Rev 1

ezurio

P/N:453-00224R

D/C:SSYYWWD

CE **UK**
CA

Q'TY:XXXXPCS

BOX_ID:BXXXXXXYMDXXXXXXX



Chat

M/N:BL54L15u Rev X

ezurio

FCC ID:SQG-BL54L15U

IC:3147A-BL54L15U



 201-250075

 R-C-EZU1-BL54L15U

The following labels are placed on the pizza box.

M/N:BL54L15u Rev 1

ezurio

P/N:453-00223R

D/C:SSYYWWD



Q'TY:XXXXPCS

BOX_ID:BXXXXXXYMDXXXXXXX



M/N:BL54L15u Rev 1

ezurio

P/N:453-00224R

D/C:SSYYWWD



Q'TY:XXXXPCS

BOX_ID:BXXXXXXYMDXXXXXXX



[Chat](#)

M/N:BL54L15u Rev X

FCC ID:SQG-BL54L15U

IC:3147A-BL54L15U

  201-250075

 R-C-EZU1-BL54L15U

ezurio



The following labels are placed on the master shipping carton.

ezurio

P/N:453-00223R

Revision:1

Description:*Module,*BL54L15u,
*Chip Antenna,*T/R

Quantity:XXXX

Date*Code:SSYYWWD

Country*of*Origin:XX

Ezurio

50*S.*Main*St.*Ste.*1100

Akron,*OH*44308

UK CA 

PO:XXXXXXXX



www.ezurio.com

ezurio

P/N:453-00224R

Revision:1

Description:*Module,*BL54L15u,
*Chip Antenna,*T/R

Quantity:XXXX

Date*Code:SSYYWWD

Country*of*Origin:XX

Ezurio

50*S.*Main*St.*Ste.*1100

Akron,*OH*44308

UK CA 

PO:XXXXXXXX



www.ezurio.com

Chat

M/N:BL54L15u Rev X

FCC ID:SQG-BL54L15U

IC:3147A-BL54L15U

201-250075

R-C-EZU1-BL54L15U

ezurio



Recommended Stencil Aperture

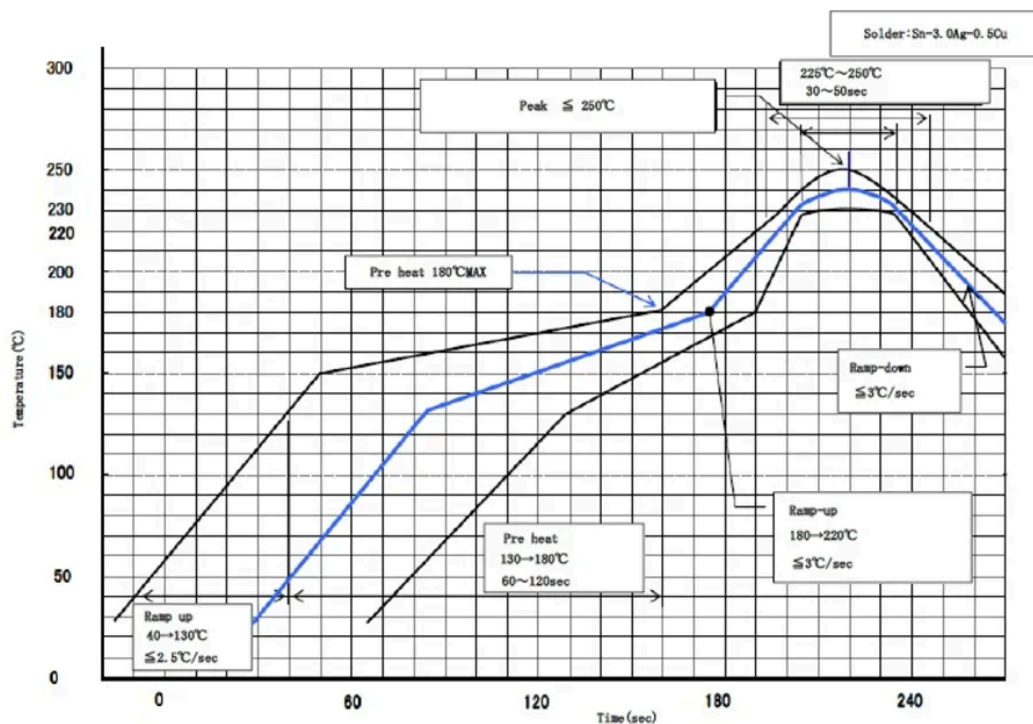
When soldering, the stencil thickness should be ≥ 0.1 mm.

Reflow Parameters/ Soldering

Convection reflow or IR/Convection reflow (one-time soldering or two-time soldering in air or nitrogen environment)

Measuring point – IC package surface

Temperature profile:



Ramp-up: 40-130 °C. Less than 2.5 °C/sec

Pre heat: 130-180 °C 60-120 sec, 180 °C MAX

Ramp-up: 180-220 °C. Less than 3 °C/sec

Peak Temperature: MAX 250 °C

225 °C ~ 250 °C, 30 ~ 50 sec

Ramp-down: Less than 3 °C/sec

Cautions when Removing the SIP from the Platform for RMA

Bake the platform PCBA before removing the BL54L15 μ module from the platform.

Remove the BL54L15 μ module by using a hot air gun. This process should be carried out by a skilled technician.

Recommended conditions for one-side component platform:

Set the hot plate at 280°C.

Put the platform on the hot plate for 8~10 seconds.

Remove the device from platform.

Chat

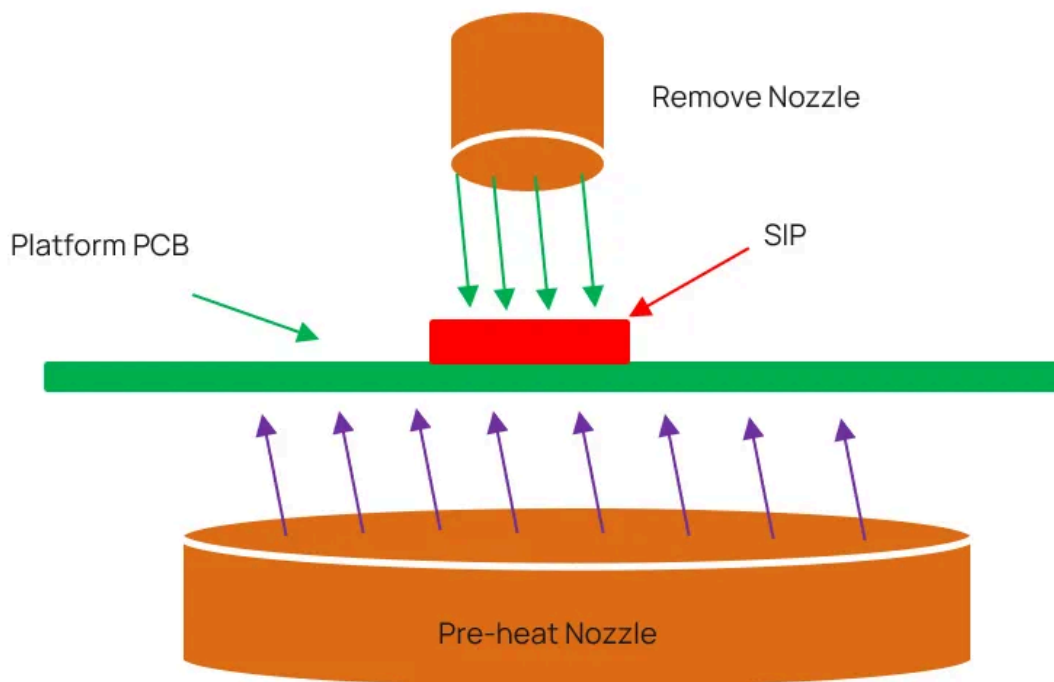


Recommended conditions for two-side components platform:

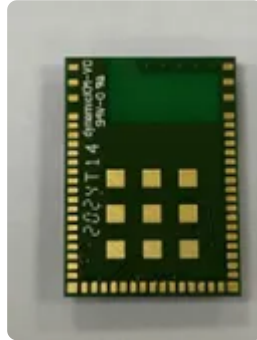
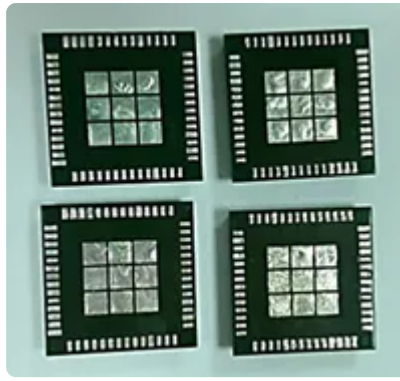
Use two hot air guns.

On the bottom, use a pre-heated nozzle (temp setting of 200~250°C) at a suitable distance from the platform PCB.

On the top, apply a remove nozzle (temp setting of 330°C). Heat until device can be removed from platform PCB.



Remove the residue solder under the bottom side of device. (Note: Alternate module pictured as an example)



(Not accepted for RMA)

(Accepted for RMA analysis)

Remove and clean the residue flux as needed.

Precautions for Use

Opening/handing/removing must be done on an anti-ESD treated workbench. All workers must also have undergone anti-ESD treatment.

The devices should be mounted within one year of the date of delivery.

The BL54L15μ modules are MSL 4 rated.

Environmental and Reliability

Environmental Requirements

Required Storage Conditions

Prior to Opening the Dry Packing

The following are required storage conditions

prior

to opening the dry packing

:

Normal temperature: 5~40 °C

Normal humidity: 80% (Relative humidity) or less

Storage period: One year or less

i Note:

Humidity means relative humidity.

After Opening the Dry Packing

The following are required storage conditions

after

opening the dry packing

(to prevent moisture absorption):

Storage conditions for one-time soldering:

Temperature: 5-25°C

Humidity: 60% or less

Period: 72 hours or less after opening

Storage conditions for two-time soldering

Storage conditions following opening and prior to performing the 1st

reflow:

Temperature: 5-25°C

Humidity: 60% or less

Period: A hours or less after opening

Storage conditions following completion of the 1st

reflow and prior to performing the 2nd

reflow

Temperature: 5-25°C

Humidity: 60% or less

Period: B hours or less after completion of the 1st reflow

Note:

Should keep A+B within 72 hours.

Temporary Storage Requirements after Opening

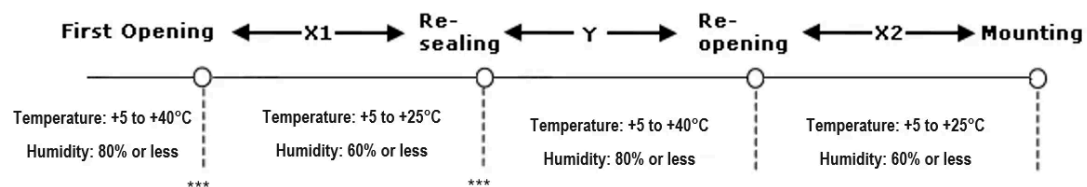
The following are temporary storage requirements after opening:

Only re-store the devices once prior to soldering.

Use a dry box or place desiccant (with a blue humidity indicator) with the devices and perform dry packing again using vacuumed heat-sealing.

The following indicate the required storage period, temperature, and humidity for this temporary storage:

Storage temperature and humidity:



*** - External atmosphere temperature and humidity of the dry packing

Storage period:

X1+X2 – Refer to Material handling information

Required Storage Conditions. Keep is X1+X2 within 72 hours.

Y – Keep within two weeks or less.

Baking Conditions

Baking conditions and processes for the module follow the J-STD-033 standard which includes the following:

The calculated shelf life in a sealed bag is 12 months at <40°C and <80% relative humidity.

Once the packaging is opened, the SiP must be mounted (per

MSL4/Moisture Sensitivity Level 4

) within 72 hours at <30°C and <60% relative humidity.

If the SiP is not mounted within 72 hours or if, when the dry pack is opened, the humidity indicator card displays >10% humidity, then the product must be baked for 48 hours at 125 °C (±5 °C).

Reflow Profile

Reliability Tests

Environmental and Mechanical

Climatic and Dynamic

Climatic and Dynamic Reliability Test Results for BL54L15u Modules

Test Item
Specification
Standard
Test Result
Thermal Shock
Temperature: -40 ~ 105°C Ramp time: Less than 10 seconds. Dwell Time: 10 minutes Number of Cycles: 350 times
*JESD22-A106 *IEC 60068-2-14 for dwell time and number of cycles
PASS
Vibration
Non-Operating Unpackaged device
Vibration Wave Form: Sine Waveform Vibration frequency / Displacement: 20-80 Hz/1.5mm Vibration frequency / Acceleration: 80-2000 Hz/20g Cycle Time: 4 min/cycle Number of Cycles: 4 cycle/axis Vibration Axes: X, Y and Z (Rotate each axis on vertical vibration table)
JEDEC 22-B103B (2016)
PASS
Mechanical Shock
Non-Operating Unpackaged device
Pulse shape: Half-sine waveform Impact acceleration: 1500 g Pulse duration: 0.5 ms Number of shocks: 30 shocks (5 shocks for each face) Orientation: Bottom, top, left, right, front, and rear faces
JEDEC 22-B110B.01 (2019)
PASS

Reliability Prediction

Ezurio Part Number
Environment
Standard
Test Result 45 °C (Hours)
453-00223R 453-00223C
Ground, Fixed, Uncontrolled
Telcordia Issue 3
6,126,139
453-00224R 453-00224C
Ground, Fixed, Uncontrolled
Telcordia Issue 3
6,858,984
453-00223R 453-00223C
Mobile, Fixed, Uncontrolled
Telcordia Issue 3
2,297,302
453-00224R 453-00224C
Mobile, Fixed, Uncontrolled
Telcordia Issue 3
2,572,119
Ezurio Part Number
Environment
Standard
Test Result 105 °C (Hours)
453-00223R 453-00223C
Ground, Fixed, Uncontrolled
Telcordia Issue 3
639,566
453-00224R

453-00224C
Ground, Fixed, Uncontrolled
Telcordia Issue 3
724,666
453-00223R 453-00223C
Mobile, Fixed, Uncontrolled
Telcordia Issue 3
239,837
453-00224R 453-00224C
Mobile, Fixed, Uncontrolled
Telcordia Issue 3
271,750

Regulatory, Qualification & Certifications

Regulatory Approvals

Full regulatory information on the BL54L15u, including the Regulatory Information Guide, grants, and test reports are available on the [BL54L15u product page](#).

The BL54L15u holds current certifications in the following countries:

Country/Region
Regulatory ID
USA (FCC)
SQG-BL54L15U
EU (ETSI)
N/A (No ID Number Required)
UKCA
N/A (No ID Number Required)
Canada (ISED)
3147A-BL54L15U
Japan (MIC)
201-250075
Australia (RCM)
N/A
New Zealand (RCM)
N/A
Korea
R-C-EZU1-BL54L15U

Certified Antennas

Manufacturer
Model
Ezurio Part Number
Type
Connector
Peak Gain @ 2400-2500 MHz
Peak Gain @ 2400-2480 MHz
Ezurio (Laird Connectivity)
NanoBlue
EBL2400A1-10MH4L
PCB Dipole
IPEX MHF4
2 dBi
-
Ezurio (Laird Connectivity)
FlexPIFA
001-0022
PIFA
IPEX MHF4
-
2 dBi
Mag.Layers
EDA-8709-2G4C1-B27-CY
0600-00057
Dipole
IPEX MHF4
2.32 dBi
-
Ezurio (Laird Connectivity)
mFlexPIFA
EFA2400A3S-10MH4L
PIFA
IPEX MHF4
-
2 dBi

Ezurio (Laird Connectivity)
iFlexPIFA Mini
EFG2401A3S-10MH4L
PIFA
IPEX MHF4
-
2 dBi
Ezurio (Laird Connectivity)
Ezurio NFC
0600-00061
NFC
N/A
-
-

Bluetooth SIG Qualification

The Bluetooth Qualification Process promotes global product interoperability and reinforces the strength of the Bluetooth® brand and ecosystem to the benefit of all Bluetooth SIG members. The Bluetooth Qualification Process helps member companies ensure their products that incorporate Bluetooth technology comply with the Bluetooth Patent & Copyright License Agreement and the Bluetooth Trademark License Agreement (collectively, the Bluetooth License Agreement) and Bluetooth Specifications.

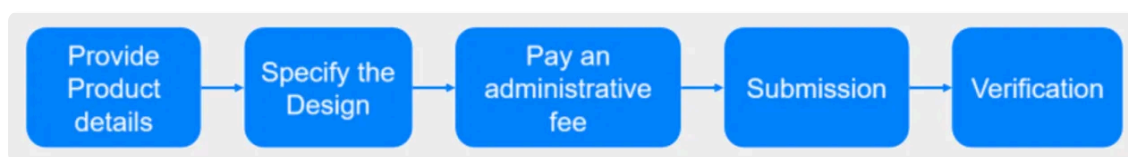
The Bluetooth Qualification Process is defined by the [Qualification Program Reference Document \(QPRD\)](#).

To demonstrate that a product complies with the Bluetooth Specification(s), each member must for each of its products:

Identify the product, the design included in the product, the Bluetooth Specifications that the design implements, and the features of each implemented specification

Complete the Bluetooth Qualification Process by submitting the required documentation for the product under a user account belonging to your company

The Bluetooth Qualification Process consists of the phases shown below:



To complete the Qualification Process the company developing a Bluetooth End Product shall be a member of the Bluetooth SIG. To start the application please use the following link if you are not already a Bluetooth SIG Member:

[Apply for Adopter Membership](#)

, (Associate Membership also available which has a yearly membership fee).

Scope

This guide is intended to provide guidance on the Bluetooth Qualification Process for End Products that reference a single existing design, that has not been modified, (refer to Section 3.2.1 of the [Qualification Program Reference Document \(QPRD\)](#)).

This option applies to a Member qualifying a Product that includes an existing Design that has a DN, QDID, or DID and that Design has not been modified (e.g., rebranding a Qualified Product from another Member). The Design identified by the DN, QDID, or DID may only implement Bluetooth Specifications that are active or deprecated at the time of Submission. No modifications may be made to the Design, including changes to the ICS Form.

Changes to the Product outside the Design are allowed, including:

Enabling technologies

Changes to the industrial design

Chat

Changes to the communication technology other than Bluetooth

Changes to the features that are unrelated to Bluetooth, branding, packaging, colour, shape, Product name, or Model number

Members are responsible for assessing that modifications to the Product outside of the Design do not affect compliance with Bluetooth Specifications or result in a change to the ICS Form.

For the purposes of this document, it is assumed that the member is combining a single unmodified Core-Complete Configuration.

Qualification Steps When Referencing a single existing design, (unmodified) – Option 1 in the QPRD

For this qualification option, follow these steps:

To start a listing, go to:

<https://qualification.bluetooth.com/>

Select

Start the Bluetooth Qualification Process

.

Product Details to be entered:

Project Name (this can be the product name or the Bluetooth Design name).

Product Description

Model Number

Product Publication Date (the product publication date may not be later than 90 days after submission)

Product Website (optional)

Internal Visibility (this will define if the product will be visible to other users prior to publication)

If you have multiple End Products to list then you can select 'Import Multiple Products', firstly downloading and completing the template, then by 'Upload Product List'. This will populate Qualification Workspace with all your products.

Specify the Design:

Do you include any existing Design(s) in your Product? Answer Yes, I do.

Enter the single DN or QDID used in your, (for Option 1 only one DN or QDID can be referenced)

Once the DN or QDID is selected it will appear on the left-hand side, indicating the layers covered by the design.

Select 'I'm finished entering DN's

What do you want to do next? Answer, 'Use this Design without Modification'

Save and go to Product Qualification Fee

Test Plan and Documentation

As no modifications have been made to the combined designs the tool should report the following message:

'No test plan has been generated for your new Design. Test declarations and test reports do not need to be submitted. You can continue to the next step.'

Save and go to Product Qualification fee

Product Qualification Fee:

Do you include any existing Design(s) in your Product? Answer Yes, I do.

Enter the single DN or QDID used in your, (for Option 1 only one DN or QDID can be referenced)

Once the DN or QDID is selected it will appear on the left-hand side, indicating the layers covered by the design.

Select 'I'm finished entering DN's

What do you want to do next? Answer, 'Use this Design without Modification'

Save and go to Product Qualification Fee

Submission:

Some automatic checks occur to ensure all submission requirements are complete.

To complete the listing any errors must be corrected

Once you have confirmed all design information is correct, tick all of the three check boxes and add your name to the signature page.

Now select 'Complete the Submission'.

You will be asked a final time to confirm you want to proceed with the submission, select 'Complete the Submission'.

Qualification Workspace will confirm the submission has been submitted. The Bluetooth SIG will email confirmation once the submission has been accepted, (normally this takes 1 working day).

Download Product and Design Details (SDoC):

You can now download a copy of the confirmed listing from the design listing page and save a copy in your Compliance Folder

For further information, please refer to the following webpage:

<https://www.bluetooth.com/develop-with-bluetooth/qualification-listing/>

Example Design Combinations

The following gives an example of a design possible under option 1:

Ezurio End Product design using Nordic Component based design

Chat

Design Name
Owner
Design ID
Link to listing on the SIG website
Core Spec Version
BL54L15/ BL54L15μ
Ezurio
Q345095
https://qualification.bluetooth.com/ListingDetails/274104
6.0
BL54L15/ BL54L15μ
Ezurio
Q387119
https://qualification.bluetooth.com/ListingDetails/334158
6.2

Qualify More Products

If you develop further products based on the same design in the future, it is possible to add them free of charge. The new product must not modify the existing design i.e add ICS functionality, otherwise a new design listing will be required.

To add more products to your design, select 'Manage Submitted Products' in the [Getting Started](#) page, Actions, Qualify More Products. The tool will take you through the updating process.

Ordering Information

Part Number
Product Description
453-00223R
Module, BL54L15μ, (Nordic nRF54L15), Chip antenna, Tape/Reel
453-00224R
Module, BL54L15μ, (Nordic nRF54L15), RF Trace Pin, Tape/Reel
453-00223C
Module, BL54L15μ, (Nordic nRF54L15), Chip antenna, Cut Tape
453-00224C
Module, BL54L15μ, (Nordic nRF54L15), RF Trace Pin, Cut Tape
453-00223-K1
Development kit, Module, BL54L15μ (Nordic nRF54L15), Chip antenna
453-00224-K1
Development kit, Module, BL54L15μ (Nordic nRF54L15), RF Trace Pin

Legacy - Revision History

Version
Date
Notes
Contributor(s)
Approver
0.1
02 Oct 2024
Preliminary release.
Raj Khatri
Jonathan Kaye
0.2
18 Oct 2024
Updated table4 external antenna list to Added antenna Ezurio iFlexPIFA Mini EFG2401A3S-10MH4L to External Antenna Integration with BL54L15μ RF trace pin variant (453-00224) Change Mag Layers EDA-8709-2G4C1-B27-CY antenna gain from 2dBi to 2.32dBi.
Raj Khatri
Jonathan Kaye
0.3
5 Nov 2024
Updated height of module to 1.75 mm.
Dave Drogowski
Jonathan Kaye
0.4
20 Nov 2024
Updated power supply voltage to 1.7 – 3.6 V.
Dave Drogowski
Jonathan Kaye
0.5
19 Feb. 25
Updated 6. Mechanical Details, 7.On -Board PCB Chip Antenna Characteristics, 8.Application Note for Surface Mount Modules, 9.Reliability Test
Louis Chang
Jonathan Kaye
0.6
18 Mar. 25
Update Power consumption in 2.1
Louis Chang

Jonathan Kaye
0.7
11 Apr. 25
Change (VDD_nRF) operating range from 1.7V-3.6V to 1.7V-3.5V. Add VDD_nRFsupply voltage (1.7V-3.3V) under extended operating Temperature (85 to 105C) Update Absolute maximum ratings of VDD_nRF (-0.3V-3.6V)
Louis Chang
1.0
26 June 2025
Initial Release.
Louis Chang Dave Drogowski
Jonathan Kaye
1.1
1 Aug 2025
Updated max output power to +8 dBm
Dave Drogowski
Jonathan Kaye
1.2
27 Nov 2025
Update average Power consumption in 2.1
Louis Chang
Jonathan Kaye
1.3
16 Dec 2025
Update Operation conditions and Absolute maximum ratings in 3.3 Adding antenna 3D pattern in On -Board PCB Chip Antenna Characteristics
Louis Chang
Jonathan Kaye
1.4
18 Dec 2025
Fixed a remaining figure to + 8 dBm power consumption.
Rikki Horrigan
Dave Drogowski